

High-Stakes Performance Thresholds, Venue Choice, and the Market for Downhill Marathons*

Adam Looney[†]

Preliminary draft.
May, 2026

Abstract

Institutions that allocate scarce slots with a common performance threshold implicitly assume performance is comparable across the environments in which it is produced. When candidates choose those environments, the threshold misallocates slots—and creates a market for favorable venues. I study this using the Boston Marathon’s qualifying system, in which one public time standard applies at any certified marathon, using 17.2 million finish times and an AKM decomposition of times into runner and course effects, validated out of sample at Boston itself. Course effects span more than 20 minutes at the qualifying margin. Three findings. First, the screening distortion is large and growing: the share of qualifiers whose qualification depends on their course roughly doubled—among actual Boston participants it now exceeds one in five—and a venue-adjusted standard would displace about 13.9% of the field, robustly 10.5–14.9% across the index suite and roughly an order of magnitude above what estimation noise alone can generate. Second, candidates respond: near-threshold runners sort to fast venues, the sorting jumps at each announced tightening, and bunching tracks the moving standard on fast courses only. Third, venues respond: purpose-built fast courses entered and expanded until venue supply became a first-order source of misallocation, of the same order as runner sorting, under every counterfactual baseline. The same forces contaminate the value-added measure itself; I show how moving-threshold placebos, retests, and directional mover asymmetries detect and bound the contamination—tools that apply wherever fixed-effect performance measures are used for high-stakes selection.

Keywords: screening, performance thresholds, venue heterogeneity, AKM decomposition, misallocation

JEL Classification: D82, J24, L83, Z20

*I am grateful to colleagues at the University of Utah for helpful comments and suggestions, and to Scott Tannler for excellent research assistance. The author has no financial relationships or other potential conflicts of interest to disclose. This research received no external funding or research support.

[†]Marriner S. Eccles Institute, David Eccles School of Business, University of Utah. Email: adam.looney@eccles.utah.edu.

1 Introduction

Many institutions allocate scarce opportunities using common merit thresholds: admission for students above a test-score cutoff, promotion for workers who hit a performance target, certification for candidates who pass a standardized benchmark. These systems are often treated as meritocratic because everyone faces the same nominal bar. But when candidates can choose the environment in which performance is measured—classes with different grading rigor, classrooms with different student composition, assignments where target attainment is easier—a common threshold confounds ability with venue choice.

What is missing from how we usually think about this problem is that it is not only candidates who respond. When a screening rule rewards favorable environments, the environments themselves become a product: someone can enter the market to supply them. The candidate side of this feedback is well documented—students sorting into easier-grading departments, teaching to the test, issuers shopping for ratings. The supply side is almost never measurable, because in most settings the “evaluation environment” is not a discrete, countable thing whose difficulty can be estimated. This paper measures both sides. Its central finding is that the supply response—purpose-built fast venues entering and expanding—grew to rival candidate sorting as a source of the misallocation produced by the screen.

The laboratory is the Boston Marathon’s qualifying system. The qualifying standard (BQ) is public, the candidate venues are observable, runners freely choose among them, and the venues enter, grow, and market themselves partly around the rule. Boston allocates scarce entries by ranking applicants on time at any recent certified marathon; persistent excess demand has driven the standard progressively tighter, and in 2025 the BAA announced that beginning with the 2027 race it will penalize qualifying times from steeply downhill courses—a principal publicly conceding that the common threshold had begun to bend the market it was meant to screen. To measure venue difficulty I gather 17.2 million finish times at over 1,000 marathon courses, link runners across races into a panel, and estimate an AKM decomposition of log finish times into runner and course effects (Abowd, Kramarz and

Margolis, 1999; Card, Heining and Kline, 2013). Estimated course effects span more than 20 minutes for runners at the qualifying threshold, while the standard has historically been identical everywhere. The index is validated out of sample at the prize venue itself: when the same runner subsequently runs Boston, a course estimated to be 1% faster predicts a commensurately slower Boston finish ($\hat{\beta} = 0.851$ on the retest, 0.933 across all cross-race pairs), and observable terrain alone explains $R^2 = 0.83$ of the cross-course variation.

The first finding is that the screening distortion is large and growing. The share of apparent qualifiers whose qualification depends on the course they ran rose from about 9.7% in the Boston 2004 cohort to about 18.4% in the Boston 2019 cohort. Among the runners actually on the Boston start line—the people occupying the scarce slots—the rate is higher still: more than one in five (22.1% in the 2019 cohort, 23.2% by 2026), because fast-course qualifiers claim their slots at above-average rates. Holding the field size fixed, a venue-adjusted standard would displace about 13.9% of recent qualifiers (bootstrap 95% CI 13.1–13.5%) in favor of faster runners from slower courses. The displacement is not estimation noise: an index estimated with the ranked cohorts excluded reproduces it (13.2%), course effects of pure noise generate only 1.7%, and the median displaced runner was 3.0 minutes inside the standard. The displaced field would also be slower at Boston itself: replacement qualifiers are predicted to run about 7.9 minutes faster at Boston than those they displace.

The second finding is that candidates respond. Near-threshold runners select fast courses at disproportionate rates, and the gap has grown: the excess fast-course share among near-BQ finishers roughly doubled across eras. The differential also jumps at each announced tightening of the standard (difference-in-differences estimate +3.2 percentage points after the February 2011 announcement). Bunching in the finish-time density tracks the moving standard, concentrates on fast courses, and is absent at round-number placebos.

The third finding is that venues respond. The number of fast US courses expanded 2.3-fold (from 23 to 54 between 2004 and 2019), the share of US finishers at fast courses 2.3-fold (from 5.7% to 13.1%), and purpose-built Boston-qualifying products—the REVEL

series, the “BQ.2” races—entered on a timeline aligned with announced tightenings and market themselves explicitly as qualifying venues. In a decomposition of the misallocation rate, the venue-supply channel contributes 4.5 percentage points by the Boston 2019 cohort (3.2–5.1 across counterfactual baselines and orderings), alongside runner-side sorting of 6.6 points (5.5–7.8). The screening rule does not merely rank candidates imperfectly; it creates a market in which qualifying performances are produced.

The AKM model is a measurement device, not a causal model of course choice, and the exogenous-mobility assumption fails here *by design*: the threshold induces exactly the sorting the paper studies. The paper’s response is to decompose the observed index—into physical venue speed, selection into venue, target effort and peaking, and order effects—and to sequence the argument so that the decomposition is needed only where it matters. The demand and supply evidence uses a coarse fast/slow classification that every correction leaves intact; the accounting that depends on course-by-course magnitudes is corrected component by component (Section 8); and the headline displacement is robust (10.5–14.9%) across differently-corrected indices, so no conclusion hinges on recovering a unique “true” index.

This paper relates to several literatures. On candidates gaming a common measure: Sabot and Wakeman-Linn (1991) document students sorting into easier-grading departments, Bagues, Sylos-Labini and Zinovyeva (2008) the supply-side response of university grading standards, and Jacob (2005), Neal and Schanzenbach (2010), and Lazear (2006) the distortions of test-based accountability; Diamond and Persson (2016) and Dee et al. (2019) show that even the graders of high-stakes tests manipulate scores at salient cutoffs; the equilibrium feedback documented here is a concrete instance of Goodhart’s law (Campbell, 1979; Goodhart, 1984). On measurement: the paper builds on the AKM decomposition (Abowd, Kramarz and Margolis, 1999; Card, Heining and Kline, 2013) with Empirical Bayes shrinkage (Morris, 1983; Andrews et al., 2008), the leave-out variance framework of Kline, Saggio and Sølvesten (2020), the forecast-bias and validation logic of Chetty, Friedman and Rockoff (2014*a,b*), the sorting critique of Rothstein (2010), and the bunching estimators of

Saez (2010) and Kleven and Waseem (2013). On markets for favorable evaluation: Bolton, Freixas and Shapiro (2012) model issuer shopping among credit-rating agencies, and Dranove et al. (2003) show how published report cards induced providers to select patients—the closest empirical relative of the feedback loop measured here. On mechanism design, Barlevy and Neal (2012) characterize ordinal, within-environment incentive schemes that are robust to exactly the comparability problem this paper documents; Section 8 evaluates that class of rule for Boston.

The paper proceeds as follows. Section 2 describes the setting, data, and framework. Section 3 builds the course-speed index, and Section 4 asks, assumption by assumption, whether it should be believed—a worked example of the person-plus-environment measurement toolkit, ending with the assumption that fails. Sections 5 and 6 document the behavior behind that failure: candidate demand and venue supply. Section 7 measures what the threshold misallocates, decomposes the growth, and prices the reallocation under a venue-adjusted standard. Section 8 builds the corrected index behind that accounting and evaluates the rule classes. Section 9 draws the lessons.

2 Institutional Setting, Data, and Framework

2.1 The Boston Marathon Qualifying System

The Boston Marathon is the world’s oldest annual marathon and one of its most prestigious. Unlike most large-city marathons, Boston relies on time-based qualification rather than a lottery: runners must achieve an age- and gender-specific qualifying time at a certified marathon within a designated window. The threshold is high-stakes, clearly stated, and publicly known; as B.A.A. President Jack Fleming puts it, runners “earn a spot on the starting line” (Boston Athletic Association, 2025*b*).

Regular increases in applications have made Boston substantially more selective. For men aged 18–34 the posted standard tightened from 3:10 (through Boston 2012) to 3:05 (2013–

2019), then 3:00 (2020–2025), and now 2:55 (2026–27). Two dates matter for each tightening and the paper uses both. The *announcement* is the information event: on February 16, 2011, the BAA announced both the rolling fastest-first registration process (effective for the 2012 race) and the five-minute tightening of all standards (effective for 2013); the September 2018 and September 2024 announcements followed the same pattern. The *regime cutover* is the first race date whose qualifying window falls under the new standard (mid-September 2011, 2018, and 2024); exhibits that classify qualifying performances by the standard they faced use the cutover, while behavioral event studies use the announcement, since that is when runners could begin to reoptimize.¹

The qualifying system applies the same time cutoff across all certified marathons regardless of course difficulty: a 3:00 at the steeply downhill REVEL Big Bear and a 3:00 at the hilly Marine Corps Marathon are treated identically, though Section 3 estimates they reflect underlying performance differences of more than 20 minutes. In 2025 the B.A.A. announced, for the 2027 race, net-elevation-drop penalties (5 minutes for 1,500–2,999 feet of net drop, 10 minutes for 3,000–5,999, and disqualification at 6,000+) (Boston Athletic Association, 2025a)—the first departure from the common-cutoff rule, and, as Section 8 shows, a poorly targeted one.

2.2 Data

The data are a web-scraped panel of marathon results from race websites and aggregators (MarathonGuide.com, 2024): 17,206,046 finish-time observations at over 1,000 marathon courses in the US and 18 other countries, 2000–2025. A multi-stage entity-resolution procedure links runners across races by name, gender, and year of birth, erring toward conservative matching because false merges create spurious mobility links between courses while false splits only discard information. AKM estimation uses runners observed at two or more

¹In oversubscribed years the *effective* cutoff is the posted standard minus an admission buffer, unknown until registration closes, because Boston admits qualified applicants in order of their margin until the field fills. The posted standard is the target runners pursue and governs the behavioral analyses; the effective cutoff governs actual admission and enters the policy counterfactuals.

courses within their geographic region (11,142,687 observations on 2,631,055 regional runner identities at 1,162 courses); restricting to exact name matches yields a Spearman correlation of 0.979 with the baseline course effects. The selection and choice analyses additionally use a geocoded US-resident subpopulation of about 1.9 million runners. Appendix B details the entity resolution, sample construction, and time-variable selection, and Table B1 summarizes the samples and where each is used.

2.3 Definitions

The *posted* BQ threshold is the announced standard by age and gender; the *effective* cutoff subtracts the year-specific admission buffer. An *apparent qualifier* meets the posted standard; an *actual entrant* runs Boston in the following one or two years. The *neutral BQ margin* is the standard minus the runner’s finish time after stripping the AKM course effect, so positive values mean the runner would qualify on a Boston-equivalent course. A *fast* course is at least 2% faster than Boston on the AKM log scale (about four minutes at a 3:20 pace); *very fast* is at least 4%. “Near-BQ” means close to the runner’s own standard; exact bandwidths vary by exercise and are collected, with each exercise’s rationale, in Appendix B (Table B2).²

The setting is a concrete instance of a general screening problem, and the paper uses both vocabularies interchangeably:

²Two global conventions hold throughout. Eras are defined by the standard in force for the relevant Boston cohort (pre-2013 / 2013–2019 / 2020+ for cohort exhibits) or by the announcement cutovers (pre/post September 2011 and 2018) for exhibits that classify qualifying performances by regime; each exhibit states which. COVID-era observations are handled uniformly: the cancelled Boston 2020 cohort is shaded or omitted, and the atypical Boston 2021 field is suppressed from take-up calculations.

General object	Marathon counterpart
Agent / candidate	Runner
Evaluation environment / venue	Marathon course
Principal	Boston Athletic Association
Common threshold	BQ standard (age-gender qualifying time)
Venue supplier	Race organizer
Scarce slot	Boston entry
Venue value-added	Course speed effect

2.4 Conceptual Framework

Consider an agent with baseline performance $a \sim F(a)$ who chooses effort $e \geq 0$ and an evaluation environment $v \in \mathcal{V}$, with observed performance

$$y = a + \kappa(v) + e + \varepsilon,$$

where $\kappa(v)$ is the environment’s systematic advantage. The agent incurs cost $C(e) + D(v)$ and a principal applies a common threshold: the agent qualifies iff $y \geq \bar{y}$ (cf. Baker, 1992). Three implications are immediate and the paper documents each in turn: a common threshold confounds person and place, so equally capable agents qualify at different rates solely because of where they are evaluated; tightening the threshold amplifies venue differences, because as $\bar{y}$ moves into the thin right tail of F a fixed difference in κ shifts more probability mass across the cutoff; and the strategic response concentrates among marginal candidates, who have the most to gain from a high- κ venue.

The framework identifies two linked distortions. The *screening distortion* arises whenever a common threshold is applied to heterogeneous venues, even absent strategic behavior. The *market distortion* is the endogenous response: candidates redirect venue choices, and venues themselves enter and grow, to exploit the advantage. The first does not require the second,

but the second is induced by the first.

The supply side has an entry condition. A prospective venue pays a fixed cost to enter, offers advantage κ , and earns margins on the candidates it attracts. Its customers are the candidates for whom κ has option value: those within reach of $\bar{y}$ only at a favorable venue. Each tightening strands a dense band of aspirants just outside the new standard. That raises the mass of candidates for whom a high- κ venue restores qualification, so the return to entry rises exactly when the standard tightens: purpose-built high- κ venues should enter after announced tightenings. And because the rule rewards κ however produced, entrants will produce it in whatever form the rule does not police. Section 6 tests the timing prediction; Section 8 returns to the form-shifting one.

The same framework disciplines measurement. What an AKM regression recovers for each course is the *observed* venue value-added,

$$\psi_c^{\text{naive}} = \underbrace{\delta_c}_{\text{physical venue speed}} + \underbrace{S_c}_{\text{selection into venue}} + \underbrace{E_c}_{\text{target effort / peaking}} + \underbrace{O_c}_{\text{order / edge effects}} + \text{noise}, \quad (1)$$

because the runner effect absorbs each runner’s *average* state but not the systematic within-runner effort, peaking, and ordering that correlate with venue choice. The component magnitudes are measured, not assumed (Section 8). One normative sentence guides the policy analysis: a venue-adjusted standard should neutralize δ_c —gravity is not merit—but not effort, since training to peak at a goal race is what the institution means to reward.

3 Measuring Course Speed

This section builds the measurement tool: a course-speed index estimated on the full panel. Section 4 then asks, assumption by assumption, whether the tool should be believed—and finds that the one assumption that fails does so in a way that is itself the subject of the paper.

3.1 Estimator

I estimate

$$\ln T_{irt} = \alpha_i + \delta_r + \gamma_t + f(\text{age}_{it}, \text{male}_i) + g(W_{rt}) + \varepsilon_{irt}, \quad (2)$$

where T_{irt} is the finish time of runner i in race r in year t , α_i is a runner effect, δ_r a course effect, γ_t a year effect, f an age–gender polynomial, and g a quadratic in demeaned race-day temperature. Identities are defined within five geographic regions to limit false links; estimates are normalized to Boston ($\hat{\delta}_{\text{Boston}} = 0$) using about 24,000 bridge runners, shrunk by Empirical Bayes, and published for the 1057 courses with at least 50 observations. Appendix A gives the full specification, the progressive R^2 decomposition, the shrinkage estimator, and an algebraic invariance result (the fixed-field displacement of Section 7 is invariant to any common shift of the index, so only cross-course *differential* error can matter for the headline); Appendix C reports sample-restriction robustness.

3.2 What the Index Shows

Figure 1 shows the result. Purpose-built downhill courses run 4–9% faster than Boston; major city races and trail marathons run slower; estimated effects span more than 20 minutes for runners at the qualifying margin (the full published index is Appendix G). Most courses are *slower* than Boston, so most BQ pursuit happens at a venue disadvantage—a fact Section 7 returns to when it prices the reallocation.

Two facts anchor the index to physical reality. Observables: terrain, surface, and climate explain $R^2 = 0.83$ of the cross-course variation; gross elevation gain is the dominant predictor ($R^2 = 0.73$ alone), while net elevation drop—the proxy the BAA would later choose—is nearly irrelevant ($R^2 = 0.05$; Table A4). Stability: the median within-course year-to-year standard deviation of course means is 2.8 minutes, small against a cross-course range of more than 40 minutes (Appendix D). The sharpest validation—the same runners retested across courses—belongs to the next section, because it is also the test of the model’s central

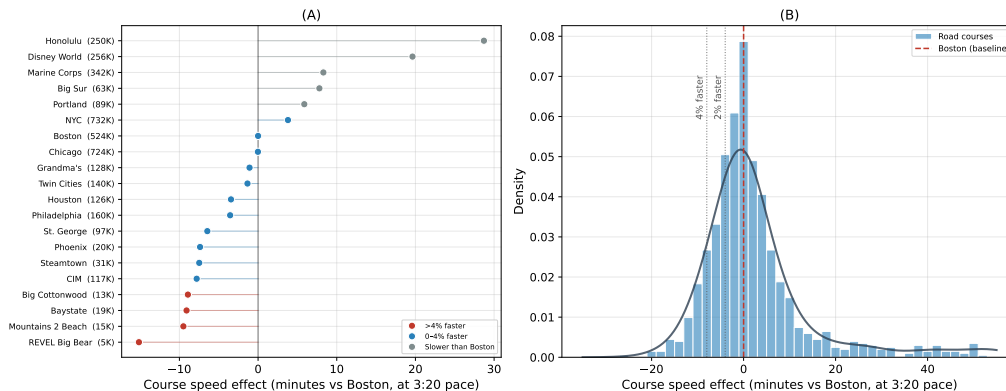


Figure 1: Course-speed distribution and benchmark courses (AKM course effects vs. Boston). Panel A: benchmark marathons in minutes relative to Boston. Panel B: the full distribution for road marathons.

assumption.

4 Should the Measurement Be Believed? Three Questions

Equation (2) is the workhorse of modern measurement economics: the same additive person-plus-environment decomposition as AKM wage models (Abowd, Kramarz and Margolis, 1999; Card, Heining and Kline, 2013) and teacher value-added (Chetty, Friedman and Rockoff, 2014a), with runners as workers and courses as firms. Everything the device reports rests on three assumptions. In most applications they must be defended indirectly, because the environments are hard to characterize and the movers are few; here each can be put to the data directly. This section asks the three questions in turn. Two receive unusually clean yeses. The third receives a no—and the anatomy of that no organizes the rest of the paper.

4.1 Is Additivity Good Enough?

Additive separability says a course shifts every runner's log time by the same amount. It is the assumption every fixed-effects application waves through, because testing it requires

observing the same person in many environments; here, millions of repeat-runner pairs test it directly. For each same-runner pair of races, the change in log finish time should equal the difference in the two courses’ estimated effects, one for one. It does: regressing within-runner time changes on course-effect differences gives $\hat{\beta} = 0.933$ across all 10,934,525 cross-race pairs, and the Boston retest—pairing each Boston run with the same runner’s prior marathons within 18 months, so that the index estimated elsewhere is tested out of sample at the prize venue—gives $\hat{\beta} = 0.851$ on 690,058 pairs (Figure 2; Appendix A, Table A2). Nor does the validation rest on Boston-destination pairs, the most campaign-selected population in the data: rerunning the retest with lottery-entry majors as the destination gives slopes of 0.99 (Chicago), 1.00 (New York), and 0.83 (Marine Corps) (Appendix Table A5), so the transfer relationship is not an artifact of the Boston-bound population alone.

The slopes sit close to one but not on it, and the deviations are themselves diagnostic—a first taste of the paper’s central theme, that performance measures absorb the behavior and the record-linkage around them. Stratifying the pairs by how many distinct people share a runner’s name separates two contaminations with opposite signs (Appendix Table A3). When entity resolution accidentally merges two different people who share a name, the resulting “mover” is not one person running two races but two people compared across venues—and because faster runners are over-represented at fast courses, the fused pair typically sets a faster person’s time at the faster course against a slower person’s time at the slower one. The measured gap then overshoots the course difference: among common names, where such fusions concentrate, the slope rises *above* one (1.16 pooled, 1.36 into Boston)—selection on ability masquerading as movement, the linkage analog of the sorting this paper studies. Restricting to medium-frequency names with every low-confidence link excluded brings the slope to 0.98 (0.93 into Boston). The second contamination runs the other way: slopes into Boston sit below slopes elsewhere throughout the table, exactly where qualifying-campaign peaking predicts under-transfer—performances peaked at the origin race do not carry fully into the goal race, behavior Section 8 measures and corrects. Additivity survives

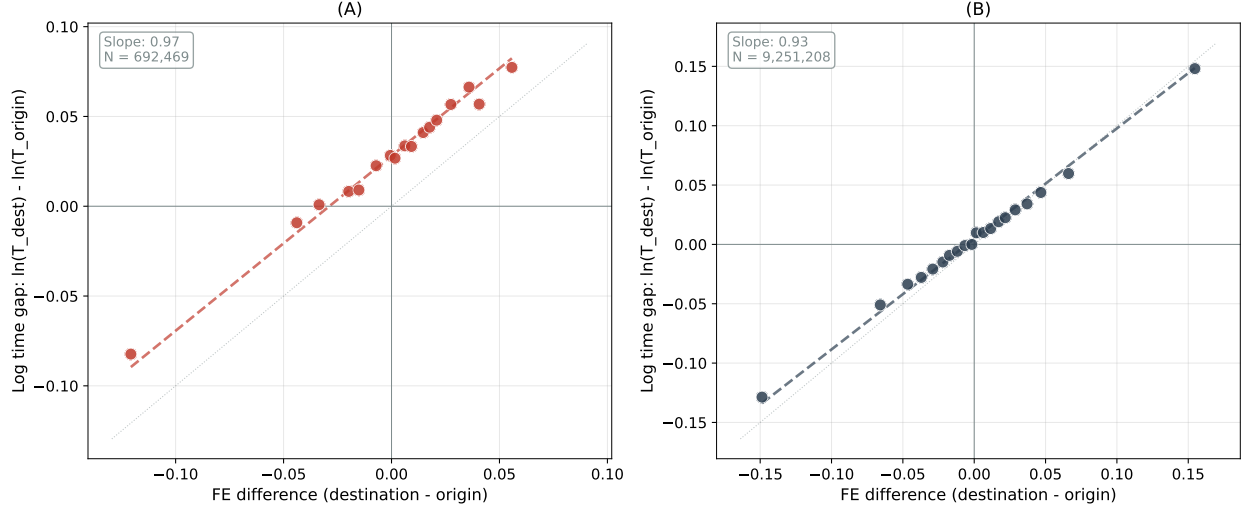


Figure 2: Course effects are additively separable across race pairs. Panel A: Boston as a standardized retest—log time gap vs. FE difference for Boston-destination pairs. Panel B: all cross-race pairs. Each point is a quantile bin mean; dashed line is the fit through microdata; dotted line is the 45-degree perfect additive-separability benchmark; estimates in Table A2.

as a good approximation; the deviations around it are signed, interpretable, and put to work below.

4.2 Is There Enough Mobility?

The second assumption is that the data contain enough movement between environments to separate person from place. In firm applications this is the binding constraint: identification travels only through movers, the mover graph is sparse, and limited-mobility bias inflates the dispersion of estimated firm effects (Kline, Saggio and Sølvssten, 2020). The marathon graph is different in kind: 2,631,055 runners observed at multiple courses connect 1,162 courses—roughly 2,264 movers per course, where firm-pair cells in matched employer-employee data are often single digits. Sparsity survives at the runner level (the median runner appears in only a few races), which is why the paper makes no variance-decomposition claims, shrinks each course effect by Empirical Bayes using its estimated sampling variance, and carries a conservative index that zeroes every course statistically indistinguishable from Boston (Appendix Table C7); the sampling variances are not leave-out variances in the sense of

Kline, Saggio and Sølvesten (2020), which is one reason the displacement headline is re-checked with leave-out estimation, a pure-noise placebo, and bootstrap intervals in Section 7, and grouping sparse courses in the spirit of Bonhomme, Lamadon and Manresa (2019) is a natural further discipline. One genuine limitation is a moving target rather than a sparse graph: a course effect is a property of a course-as-run, marathons occasionally re-route, and young purpose-built courses have short histories; the within-course stability diagnostic (Appendix D) bounds how much this matters on average.

4.3 Are the Moves Exogenous? No

The third assumption is that movement between environments is unrelated to the *transitory* component of performance. Stating the estimand precisely shows what is and is not at risk. The policy object is δ_c , physical venue speed: the expected log-time shift for a runner in a fixed reference state. What the regression recovers is ψ_c^{naive} in equation (1): physical speed plus the venue-specific average of behavioral state among attempts there. Because the runner effect absorbs each runner’s *average* state, persistent sorting—strong runners favoring fast courses—does no damage at all. What loads onto the course term is *differential* composition of transitory state across venues. And the qualifying threshold manufactures exactly that: runners choose fast courses for the attempts when they are trained, tapered, healthy, and one near-miss short of their standard.

This is the disagreement at the center of the teacher value-added literature, live and visible. Rothstein (2010) showed that selection on transitory shocks can contaminate fixed-effect quality measures even when selection on persistent ability is absorbed; Chetty, Friedman and Rockoff (2014a) responded with forecast-bias tests built on quasi-experimental variation. The marathon adjudicates the debate in miniature because everything each side must ordinarily assume is observable here: the incentive is a public threshold that moves on announced dates, the venue menu is discrete and physically measurable, the most relevant transitory shock (race-day weather) is recorded, and the same runner is observed on both

sides of every venue switch. The Rothstein-style failure is real—and it is signed, local, and incentive-shaped: selection concentrates near the threshold, points toward fast venues, and intensifies with each tightening. That structure is what makes the failure detectable, and it supplies the diagnostics: bunching that tracks an administratively moved cutoff, threshold-distance interactions, and directional mover asymmetries (runners are systematically slower *into* Boston than the additive model predicts, but not on the way out; Section 8).

The failure of exogenous mobility is therefore not a nuisance to be apologized for; it is the phenomenon. Sections 5 and 6 document the behavior that breaks the assumption—candidates sorting and venues entering—and Section 8 returns to measurement, scaling corrections by each course’s share of strategic users and bounding what remains.

5 The Market Response I: Candidate Demand

Section 4 ended with an assumption that fails; this section documents the behavior that breaks it—who sorts, when, how much, and through which margin—by contrasting populations that face the qualifying incentive with populations that do not: near-threshold versus inframarginal runners, BQ thresholds versus round-number placebos, announced tightenings versus surrounding years. Every estimate uses only the coarse fast/slow course classification, which is robust to all of the index corrections in Section 8; because the estimates measure deliberately different estimands on different populations, Appendix Table C2 maps each to its object.

5.1 Sorting toward faster courses near the margin

Runners near the qualifying threshold concentrate on faster courses, and the concentration has grown. Figure 3 plots the fast-course share against the neutral BQ margin by era. The share is flat far from the threshold and rises sharply near it; the near-threshold excess grew era by era—5.8, 9.8, and 12.1 percentage points relative to each era’s field in the 3:10, 3:05,

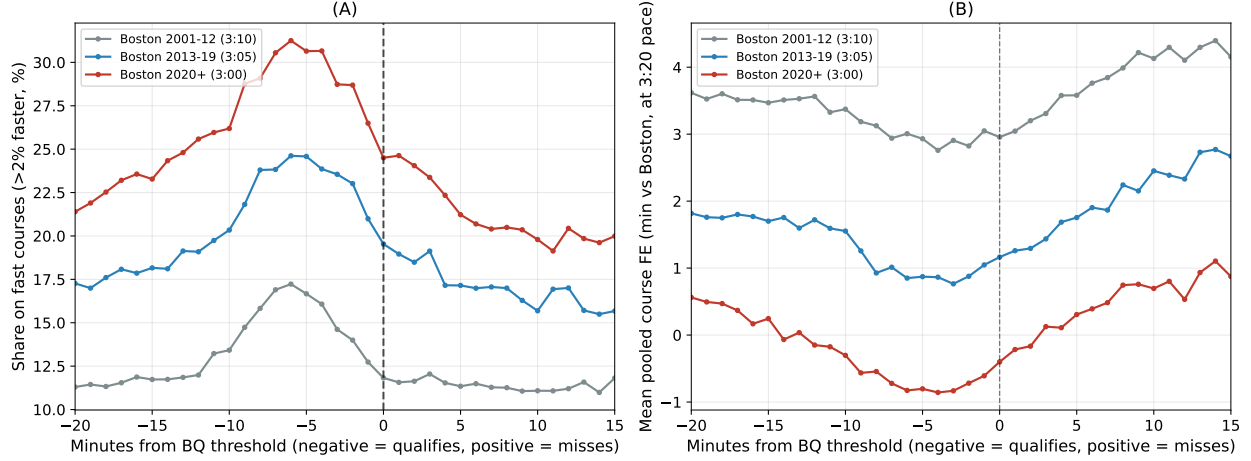


Figure 3: Fast-course share by neutral BQ margin, by era (Panel A); mean course effect by margin (Panel B). The near-threshold excess grows across eras.

and 3:00 eras. Inframarginal runners—too far from the cutoff for course choice to matter—show no comparable gradient, which addresses amenity- and geography-based alternatives: both groups face the same courses, travel costs, and calendars; only one faces the incentive.

Treating the inframarginal venue distribution as the no-incentive counterfactual, roughly one in five near-BQ runners is at a course they would not otherwise have chosen (Appendix Figure C5).

5.2 Predictable pressure and the announcement design

Selection is forecastable from pre-race information—the Chetty, Friedman and Rockoff (2014a) forecast-bias logic run in reverse, with predictable selection as the outcome of interest. Regressing pre-race BQ pressure on the chosen course’s speed gives $\hat{\theta}_\psi = -0.580$ —more pressure, faster venue—and the fast-course share peaks for runners whose prior best *just missed* the standard (Appendix Table C3).

The announced tightenings give the design-based version. In an event study of near-BQ runners’ differential fast-course choice by year, the differential is flat before the February 2011 announcement and rises steadily afterward (Figure 4); the compact difference-in-differences coefficient is +3.2 percentage points (SE 0.16) and survives home-state and year fixed effects.

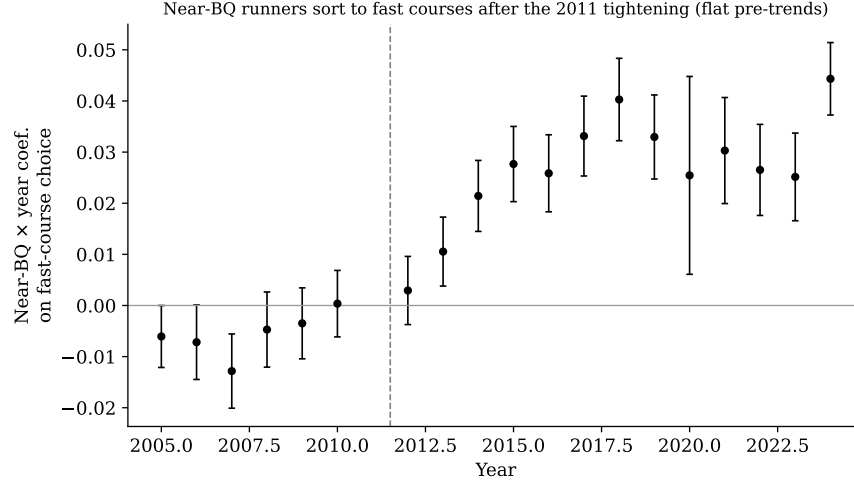


Figure 4: Event study of near-BQ runners’ differential fast-course choice, relative to 2011 (95% CIs). Flat pre-trends; sustained rise after the February 2011 announcement (the first post-announcement full qualifying year is 2012).

Because a fast course lies within 150 miles of 84% of runners, geographic access does not gate the response.

A conditional-logit model of venue selection (US runners, US courses; Appendix F) prices the same response in choice terms: utility falls in a fixed long-distance travel cost and rises in the interaction of pre-race BQ pressure with course speed. The policy-relevant translation is not the average shift in chosen-course speed—mechanically small—but the change in *qualification probability*. Switching off the pressure channel and re-solving, the model implies about -11.7 additional qualifications per 1,000 near-BQ runners, concentrated within a few minutes of the threshold, holding the course menu fixed. The model’s role is corroborative; the design-based estimates above carry the identification weight.

Qualifying standards also relax discontinuously at five-year age boundaries—a notch that in principle offers two further response margins, venue choice and attempt timing. Neither is informative here: year-of-birth data cannot classify the boundary-adjacent cells where a venue response would appear, and the attempt-retiming test returns a precise null (Appendix E). No exhibit rests on either margin.

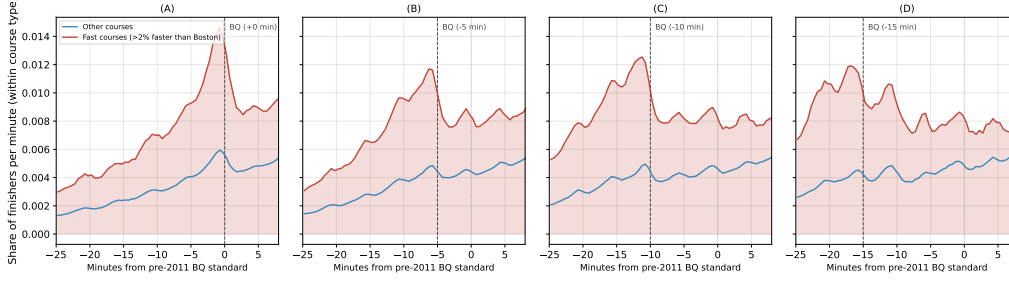


Figure 5: Bunching tracks the moving standard on fast courses and is absent on slower ones. Each panel re-centers finishes on the demographic’s pre-2011 standard; the active BQ sits five minutes further left after each tightening.

5.3 Bunching: effort and venue choice jointly

Marathon finishers bunch at salient round numbers (Allen et al., 2017). Boston-qualifying times are not round for most demographics, providing a sharp test of whether the threshold itself is a target. Three facts establish that it is, and that the response runs through fast venues. First, the bunching spike physically tracks the standard as the BAA moves it: re-centering every demographic’s finishes on its own pre-2011 standard, the fast-course density kinks exactly at the active BQ in every era, five minutes further left after each tightening, while the slow-course density stays smooth (Figure 5). A behavioral spike that follows an administratively moved cutoff cannot be a fixed feature of the environment. Second, the headline gradient: regressing each marathon’s estimated BQ-bunching premium on its course speed, the slope is -0.0131 per minute of course speed after the 2011 announcement (permutation $p < 0.001$, reassigning course speeds across marathons) versus -0.0088 before (Figure 6)—the course-speed gradient in BQ targeting is a post-tightening phenomenon. Third, polynomial excess-mass estimates by course type and era (Appendix Table C4) show the fast–slow excess-mass gradient at the BQ threshold is about three times its round-number-placebo counterpart (Appendix Figure E2); the cell-level interaction versions of these contrasts are signed identically but individually imprecise, and the marathon-level gradient above carries the inferential weight.

Finally, the sorting, bunching, and peaking facts move together when ordered by a single

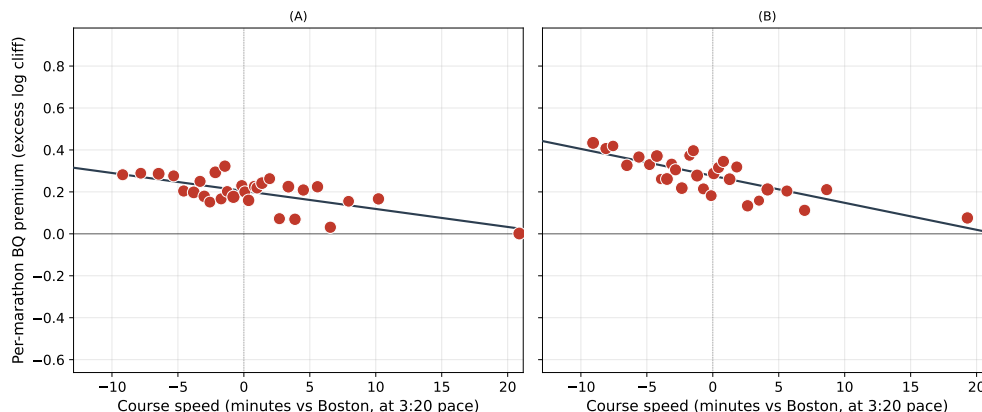


Figure 6: Per-course BQ-bunching premium by course speed, before and after the 2011 tightening: the course-speed gradient in BQ targeting steepens from -0.0088 to -0.0131 per minute of course speed.

constructed pre-race BQ-pressure index, with one exception—the peaking-by-origin cross-section is dominated by course composition—detailed in Appendix E.

The demand response has direct analogs outside sport. Sorting toward favorable venues near a threshold is the running-market version of students selecting easier-grading departments (Sabot and Wakeman-Linn, 1991), and bunching that physically tracks an administratively moved cutoff is the same signature that accountability-threshold reforms and regraded examinations provide in education (Dee et al., 2019; Diamond and Persson, 2016).

6 The Market Response II: Venue Supply

Venue entry is the side of the feedback loop least visible in prior work; this section measures it. Fast courses expanded 2.3-fold, purpose-built qualifying products entered on a timeline aligned with announced tightenings, and by the 2019 cohort the supply channel contributes 4.5 percentage points of the misallocation rate—the share of qualifiers whose qualification depends on their course, measured in Section 7—of the same order as candidate sorting.

6.1 Entry and growth of fast venues

The number of US courses at least five minutes faster than Boston at the M18–34 standard (about 3%) rose from 23 in 2004 to 54 in 2019, with the steepest growth between the 2011 and 2018 announcements (Figure 7, Panel A). The fast-course share of US finishers rose from 5.7% to 13.1% over the same period (Panel B). Entry concentrates in the product category the rule rewards: the REVEL series (purpose-built point-to-point downhill races, multiple launches 2014–2018) and the “BQ.2” races (Spring Chance, Last Chance Chicagoland, Last Chance Grand Rapids), whose branding—“BQ” is the product name—markets the screening rule itself.³

Entry is the cleanest supply margin, and Figure 8 shows its timing and composition. The fast share of newly entering US courses roughly doubles, from about 18% of entrants before 2008 to over 30% afterward, with the very-fast share roughly tripling. The flagship launches land almost entirely between the two announcements: Big Cottonwood (2012); REVEL Rockies, REVEL Canyon City, and Last Chance BQ.2 Grand Rapids (2014); BQ.2 Chicagoland and REVEL Mt. Charleston (2016); REVEL Mt. Hood, REVEL Big Bear, and Spring Chance BQ.2 (2018). Entry selection toward fast products begins around 2008, after Boston’s registration sellouts made the demand pressure visible and before the announcement that codified it—consistent with entry responding to the incentive, of which the announcement is the public signal. The consequence for the screen is summarized by one statistic: in the Boston 2019 cohort, 41% of course-dependent qualifiers—those who clear the standard only by virtue of their course—ran a course that did not exist in 2010, versus 18% of all apparent qualifiers. These are equilibrium quantities, not exogenous supply shocks: entry responds to the same incentive that moves runners, and the decomposition of Section 7.3 accordingly treats supply and sorting as accounting channels with a common upstream driver.

³The series in Figure 7 uses the five-minute definition; the 2% classification used elsewhere in the paper shows the same proportional growth.

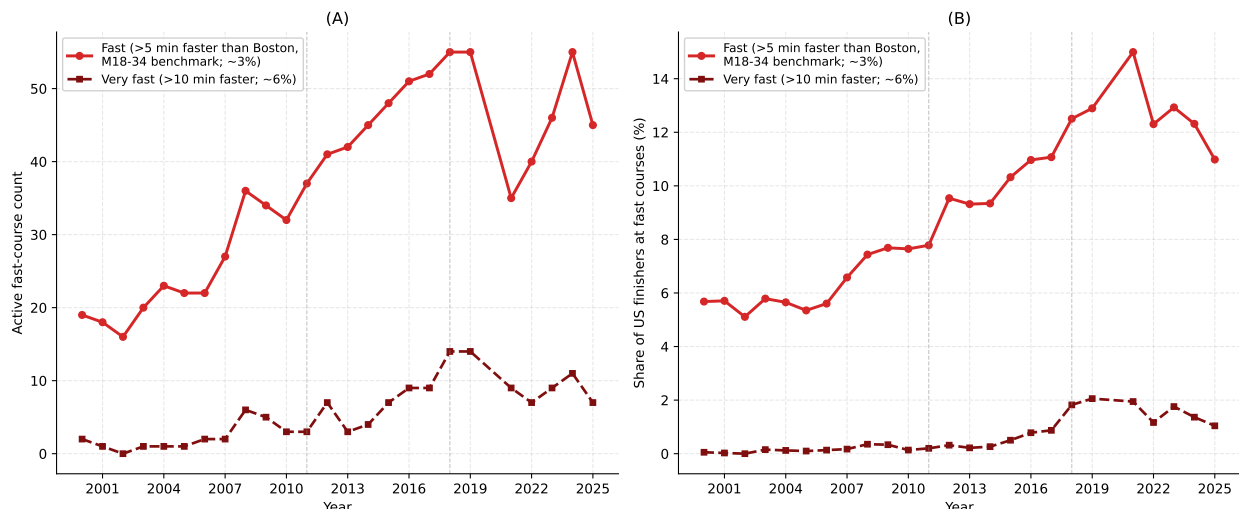


Figure 7: Growth of the fast-course market, 2000–2025. Panel A: count of US courses at least five minutes (fast) and ten minutes (very fast) faster than Boston at the M18–34 standard. Panel B: share of US finishers at such courses. Dashed lines mark the 2011 and 2018 announcements.

6.2 Which courses over-attract marginal runners

A course-level diagnostic identifies who serves the marginal demand. For each course-cohort cell, compute the share of finishers within $\pm 3\%$ of their standard, and benchmark it against a smooth in course speed: the residual is the course’s over- or under-attraction of marginal runners beyond what its speed predicts. The top over-attractors are the purpose-built “last chance” BQ events—Last Chance BQ.2 Grand Rapids, Last Chance BQ.2 Chicagoland, and Spring Chance BQ.2—whose marginal-runner shares sit roughly double what course speed alone would predict (Appendix Figure C8). The venues *created* for marginal BQ chasers are the ones the diagnostic flags.

6.3 Take-up as revealed intent

Venues created to sell qualification convert qualifiers into Boston entrants at several times the take-up rate of the majors—revealed BQ intent on the demand side, and the product working as designed on the supply side (Section 7 reports the take-up accounting). By the Boston 2019 cohort the supply contribution to the misallocation rate (4.5 pp) is of the same

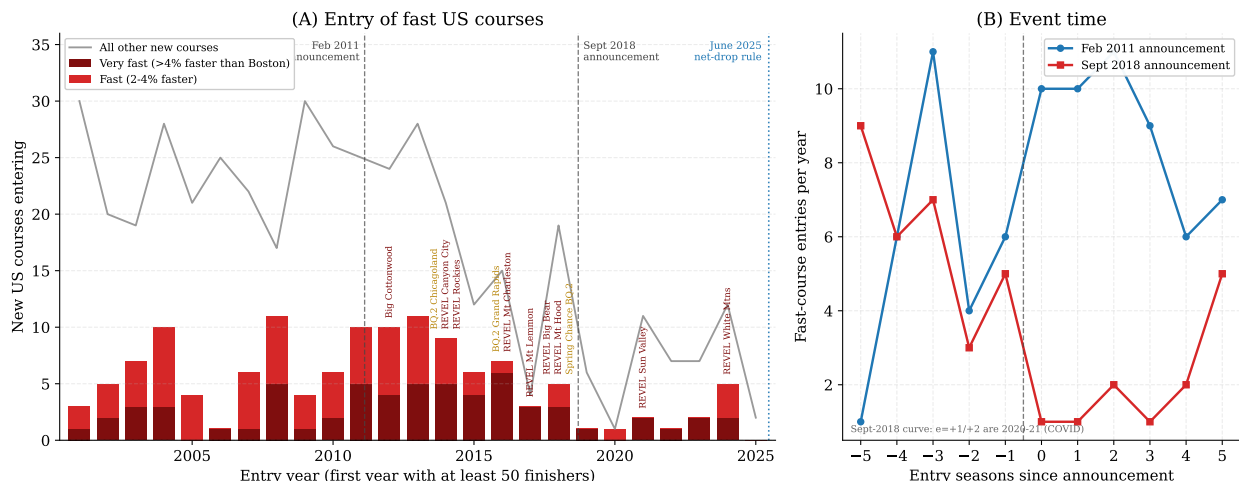


Figure 8: Entry of US road marathon courses by course-speed class, 2001–2025. Panel A: entry year and speed class, with purpose-built qualifying products labeled by name; dashed lines mark the February 2011 and September 2018 announcements and the June 2025 net-drop rule. Panel B: fast-course entries per year in event time around the two announcements. Entry year is a course’s first panel year with at least 50 finishers; courses already active in 2000 are left-censored incumbents and excluded.

order as the demand contribution (6.6 pp) under every counterfactual baseline (Section 7.3): the market distortion is two-sided.

6.4 The next round: the 2027 net-drop rule as an out-of-sample test

The BAA’s announcement of net-drop penalties for the 2027 race (June 16, 2025; effective with the qualifying window opening that September) is the loop’s next iteration, and it arrived at the very end of the sample period, so the data cannot yet show an entry response. What they can show is the target the next round will aim at. Of the 75 fast US courses active in 2024–25, 56—about 75%—fall below the 1,500-foot net-drop floor and are untouched by the rule, led by the fast-flat purpose-built products: Spring Chance BQ.2 (roughly 80 feet of net drop, a speed advantage around 16 minutes), BQ.2 Chicagoland, and the tunnel-family point-to-point races. The framework’s form-shifting prediction is that entry, course design, and marketing pivot toward this unpenalized tier; suppliers are already positioning (one

steep Utah course markets its 2026 edition as the last fast qualifier before the penalty takes effect). Section 8 develops the rule’s screening properties; the entry timeline around the 2011 and 2018 announcements (Figure 8) is the template against which the 2025 announcement will be read as the 2026–27 windows accumulate.

The supply response, too, has analogs wherever passage of a threshold is worth money or status: test-prep franchises, easy-grading course providers, diploma mills, and lenient raters are all entry on the same margin (Bolton, Freixas and Shapiro, 2012). Static audits of candidate gaming miss what may be the larger half of the equilibrium; in this setting venue supply grew to the same order as candidate sorting within a decade.

7 What the Threshold Misallocates

The market response documented in the last two sections has an allocation consequence, and this section measures it: the scale of the distortion, its growth, its decomposition into rule, supply, sorting, and take-up channels, and the counterfactual field under a venue-adjusted standard. The benchmark is the principal’s own stated objective—select the fastest runners on a common basis (“earn a spot”)—so misallocation here means deviation from that meritocratic objective, not a social-welfare claim; the BAA may also value revenue or geographic breadth, which would change the normative reading but not the positive results. All accounting in this section runs on the US road-course universe.

My primary measure is the share of apparent qualifiers whose qualification depends on the course they ran: runners with $Q_i = 1$ under the posted standard but $Q_i^* = 0$ after removing the course effect, $T_i e^{-\hat{\delta}_r} > c_{gt}$. The fast-course share of qualifiers is the intuitive secondary measure. Both are computed for three nested populations: all apparent qualifiers, and—the population that ultimately matters, because these are the people occupying scarce start-line slots—the realized Boston field.

7.1 The distortion is large and growing

Figure 9 plots the misallocation rate and the fast-course share among all finishers, among apparent qualifiers, and among Boston entrants, by Boston cohort. All series trend upward; the qualifier- and Boston-field series rise more steeply as the threshold tightens. The misallocation rate among apparent qualifiers rises from about 9.7% in the Boston 2004 cohort to about 18.4% in the Boston 2019 cohort. Among actual Boston participants the rate is higher still—22.1% in the 2019 cohort and 23.2% in the 2026 cohort—because fast-course qualifiers convert their qualifications into Boston starts at above-average rates (the take-up channel below). More than one in five runners on the Boston start line would not be there under a venue-neutral standard. The shift is visible in the terrain itself: the share of the Boston field qualifying on any net-downhill course rose from about 30 to 40%, and on courses dropping more steeply than Boston itself from 9 to 17%—runners, quite literally, racing to the bottom.

7.2 Qualifying mechanics and the shrinking menu

Two mechanical facts explain why the distortion grows with every tightening. Each runner faces a *menu* of courses at which their ability can clear the standard, and the menu contracts sharply as the standard tightens: a runner with a neutral-course baseline of 195 minutes could qualify at 172 indexed courses under the 3:10 standard, 40 under 3:05, and six under 3:00 (Appendix Figure C4, Panel A). And tightening is not neutral across venues: on slow courses the cutoff sits in the thin right tail of the ability distribution and a five-minute tightening disqualifies few runners; on fast courses—where qualification rates run roughly 30–55% at purpose-built downhill courses against low single digits at the majors—it crosses dense regions and disqualifies many (Panel B), so the fast-course share of qualifiers rises mechanically with every tightening, before any behavior.

The growth could therefore come from the rule itself, from the changing venue menu, from candidate sorting, or from who claims a slot. The decomposition separates the channels.

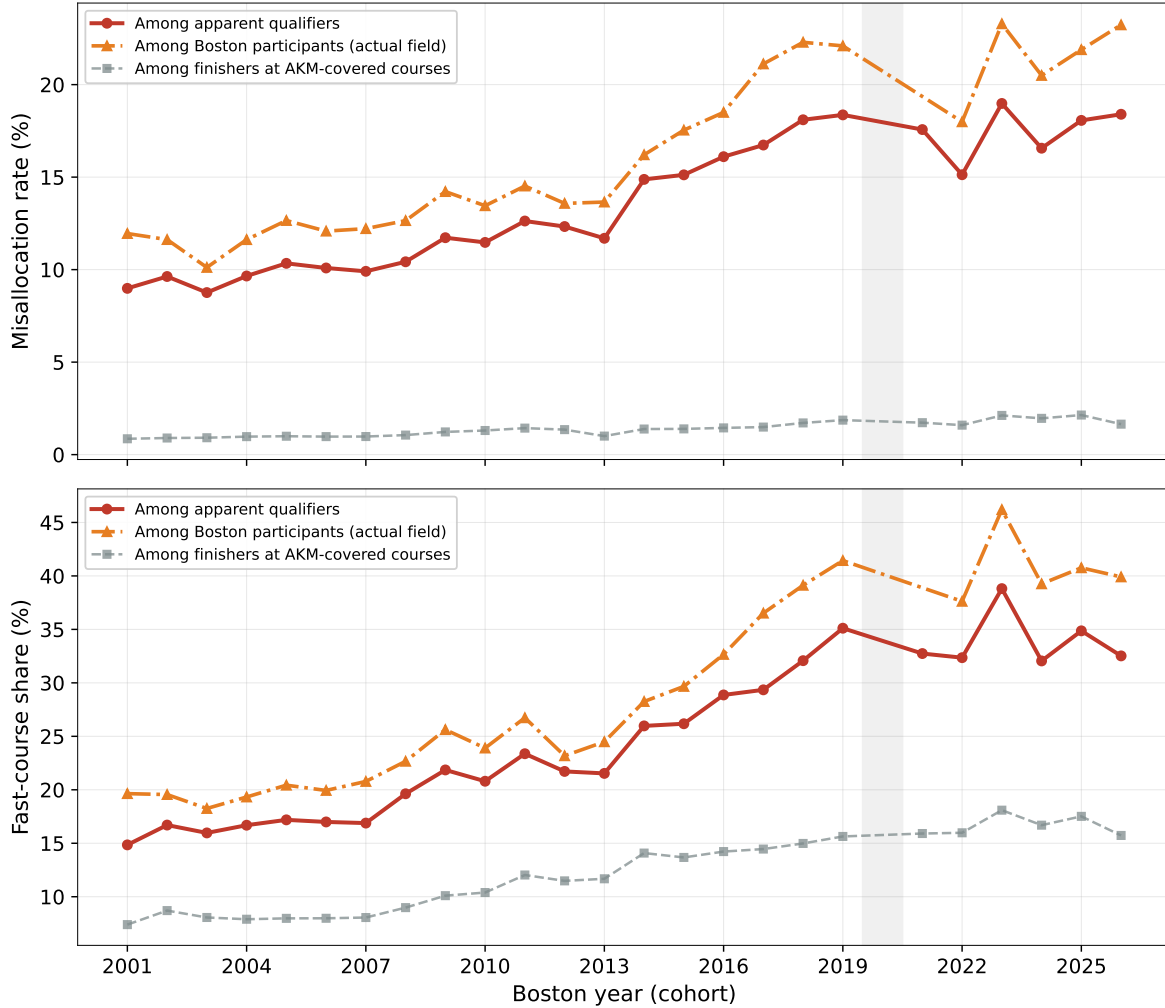


Figure 9: Misallocation rate and fast-course share among finishers, apparent qualifiers, and Boston entrants, Boston 2001–2026 cohorts. Cancelled Boston 2020 shaded; counting conventions and robustness alternatives in Appendix C.

7.3 Decomposing the growth: rule, supply, sorting, take-up

To separate the channels, I build a chain of counterfactuals that starts from random assignment of each Boston-cohort runner pool to a frozen 2005–07 course market at the 2005 standard, then progressively adds (step 2) the cohort’s actual threshold, (step 3) the cohort’s actual course menu, and (step 4) the actual assignment of runners to courses; step 5 weights qualifiers by each course’s observed take-up rate to produce the realized Boston field. The step-to-step differences are accounting contributions—threshold, venue supply, runner sorting, and differential take-up—along a chosen path with a common upstream driver, not

independently identified causal effects. Because the baseline is *random* assignment, the sorting residual captures the total deviation from random, not just its growth since the base period. The chain uses observed course-adjusted times (not estimated runner effects), so noise in the runner fixed effects does not enter the simulated qualification rates; noise in $\hat{\delta}$ enters through neutralization and is bounded by the index band of Section 8.4.

Three patterns stand out (Table 1 reports focal years; Appendix Figures C6 and C7 plot the full chains by cohort). Under random assignment to the frozen menu, misallocation is modest and roughly flat—a common threshold over a fixed venue menu is imperfect but stable. The direct rule effect contributes about 2.3 percentage points by the 2019 cohort. The venue-supply channel—applying the same random assignment to the actual, expanding course menu—contributes 5.1 points; runner sorting beyond random assignment adds 5.5 points. The two endogenous market channels together dominate the formal threshold change; Sections 5 and 6 documented each.

The random-assignment baseline is deliberately neutral, and it is not innocuous: people run their local race, so part of the step-4 deviation from random assignment is geography rather than incentive. Two alternative baselines net that out, and both *raise* the sorting channel. The first assigns each cohort’s runners to the venue distribution of its *inframarginal* runners—those more than 20 minutes from their standard on a neutral-course basis, whose venue choices embed geography and preferences but no qualification incentive. That baseline lowers the counterfactual’s fast-course tilt and moves the 2019 channels to 6.6 points of sorting and 4.5 of supply. The second—a gravity baseline with distance decay calibrated to observed travel—raises sorting further, to 7.8 points within the geocoded subsample. The direction is the opposite of the geographic intuition because full-field venue shares already contain the sorting of the marginal runners themselves: a geography-respecting counterfactual is *less* fast-tilted than the field, not more. Reordering the chain (menu before threshold) reallocates only between the threshold and supply channels. Across every baseline and ordering, sorting contributes 5.5–7.8 points and supply 3.2–5.1: candidate sorting is at least

as large as venue supply, and venue supply is never negligible (Appendix Table C5 reports the full grid).

Table 1: Misallocation decomposition at focal years (chain steps s_1 random $\rightarrow s_5$ actual field). Channels are path-dependent accounting contributions with a common upstream driver, not separately identified causal effects.

Year	Misallocation rate (%)				
	s_1 Random	s_2 +Threshold	s_3 +Menu	s_4 +Sorting	s_5 +Take-up
2004	6.3	6.3	6.3	9.7	9.7
2010	6.3	6.3	7.2	11.5	11.5
2015	6.0	7.4	10.7	15.1	15.1
2019	5.5	7.8	12.9	18.4	18.4
2024	4.1	6.7	11.7	16.6	16.6
2025	4.2	7.5	13.9	18.1	18.1

Notes: Misallocation rates in percent, US road-course qualifying universe, course effects in the Boston-state numeraire. Step 1 assigns each cohort’s runners randomly to the frozen 2005–07 course menu at the 2005 standard; steps 2–4 add the cohort’s actual threshold, menu, and assignment; step 5 weights qualifiers by observed per-course take-up. Cancelled Boston 2020 and the atypical 2021 field are excluded from take-up.

7.4 From qualification to participation: take-up

Boston admits a fixed field from qualified applicants, so the distortion in the admitted field also depends on who claims their slot. Across cohorts the average take-up rate—the share of apparent qualifiers observed at Boston within the next two years—is about 28.5% on the qualifier-weighted definition used throughout. Entity-resolution match failure deflates the level; the trend is unaffected, and the cross-course comparison is unaffected unless match rates differ by course. The main concern is composition: international-heavy fields at the majors may match at lower rates than the domestic fields of purpose-built venues, which would inflate the fast-course take-up premium. Two checks bound this. Restricting to linkage-

homogeneous US-resident qualifiers (78% of qualifiers carry a validated US residence), the fast-course premium falls from 8.4 to 5.6 percentage points and the purpose-built-venue multiple over the majors from 2.3 to 1.7—so differential match failure accounts for at most about a third of the raw premium, an upper bound since the restriction also removes genuinely lower return propensity among international qualifiers. And an external check against BAA-published per-race qualifier counts for the Boston 2025 cohort shows roughly uniform recovery across course types (68% at Chicago, the most international major, versus 71% at CIM, a domestic fast course). Take-up is consistently higher at faster courses, consistent with revealed BQ intent: purpose-built qualifying venues send their qualifiers to Boston at multiples of the rate of the majors. Step 5 of the chain shows the consequence: the fast-course share of the Boston field exceeds its share of apparent qualifiers by roughly 6–12 percentage points in most years (Figure 10). Sorting at the qualifier and take-up at Boston are two facets of the same strategic behavior.

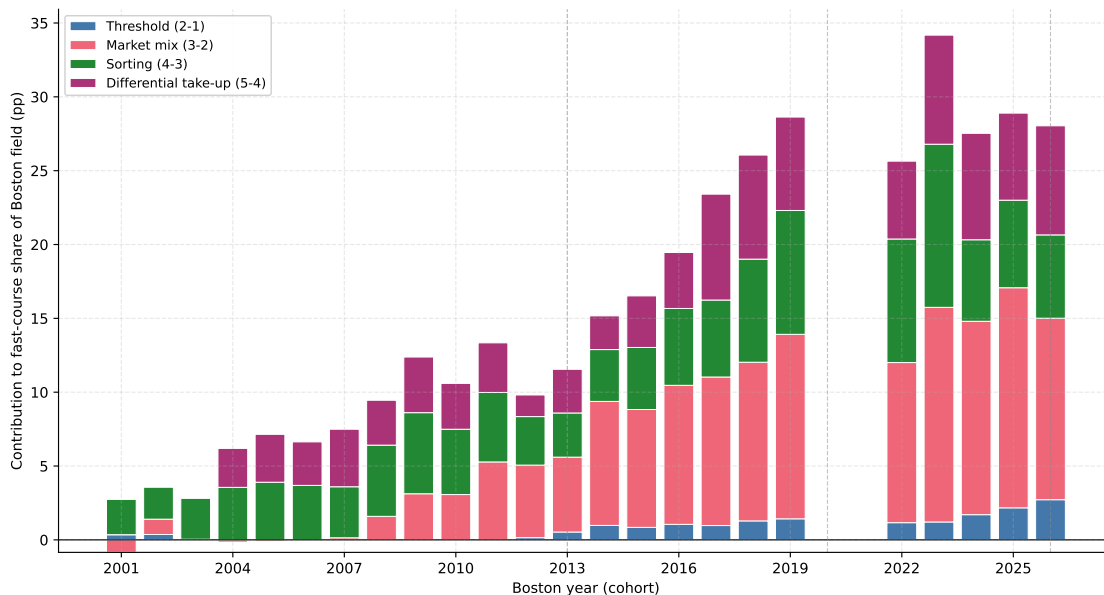


Figure 10: Decomposition of the fast-course share of the Boston field by cohort: threshold, market mix (venue supply), sorting, and differential take-up (selection into slots). COVID-era cohorts omitted.

7.5 Course-specific handicaps and the counterfactual field

Venue-adjusted standards address the screening distortion and, by removing the qualifying advantage of fast venues, would weaken both sides of the market distortion in equilibrium. The exercises here are partial equilibrium—course choices held fixed—so they measure the immediate reallocation, with an equilibrium-response bound at the end.

The natural rule adjusts each course’s qualifying times by its estimated speed: proportional handicaps, applied uniformly across age and gender groups. Most courses are *slower* than Boston, and their runners are penalized by the common standard just as surely as fast-course runners are subsidized; a venue-adjusted standard corrects both directions, while public discussion fixates on the downhill extreme and misses half the reallocation.

Holding the field size fixed and ranking all candidates by behavior-adjusted margin (the corrected index constructed in Section 8; the full suite of corrections bounds the result), 11,876 runners—13.9% of the 85,466-runner 2018–19 qualifying field (bootstrap 95% CI 13.1–13.5%)—would be displaced and replaced one-for-one. The displaced come overwhelmingly from fast courses (87.5%); the replacements come from courses at Boston’s pace or slower (Figure 11, Panel A). Displacement is 10.5–14.9% across the full index suite (Panel B; Appendix Table C6 reports every index and rule), so the conclusion survives every correction of Section 8.

Two further checks address estimation error, because ranking by $T_i e^{-\hat{\delta}_{r(i)}}$ converts noise in $\hat{\delta}$ into mechanical displacement. First, a leave-out design: re-estimating the index with 2018–19 excluded—so that no qualifying performance being ranked contributes to the course effects that rank it—moves displacement from 14.5% to 13.2% on the identical field (course-level correlation 0.991 between the two indices). Second, a pure-noise placebo: simulating course effects $\delta_c \sim N(0, \hat{s}e_c^2)$ under zero true venue heterogeneity generates displacement of only 1.7% (95% range 1.3–2.0%)—an order of magnitude below the headline. Nor is displacement knife-edge churn: the median displaced runner was 3.0 minutes inside the posted standard, only 19% were within one minute, and replacements sat a median 5.5 minutes outside it,

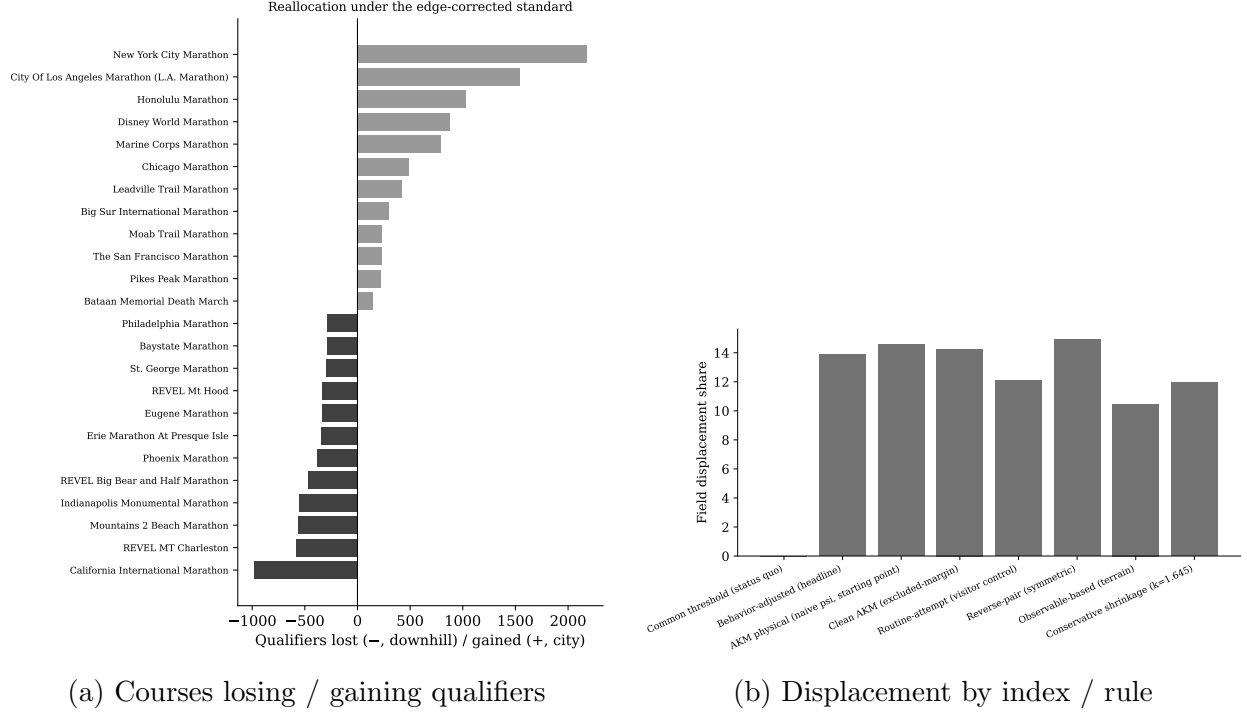


Figure 11: Reallocation under a venue-adjusted standard.

with course-neutralized times 11.9 minutes faster than the runners they displace.

The venue-adjusted field would also be faster at the prize venue. Using each runner’s permanent ability (runner effect plus age–gender profile) to predict Boston-day performance, replacement qualifiers are predicted to run about 7.9 minutes faster at Boston than the runners they displace (Figure 12)—the displaced qualify on venue, not on ability.

8 Fixing the Measure and the Rule

The displacement accounting of Section 7 ranked runners by a behavior-adjusted index; this section builds it. The naive index is corrected component by component using the decomposition of equation (1), the corrections are bounded by a suite of independently constructed indices, and the section closes with the implementable rule classes and the equilibrium response each invites.

The organizing result is proportionality. The behavioral components of ψ_c^{naive} enter

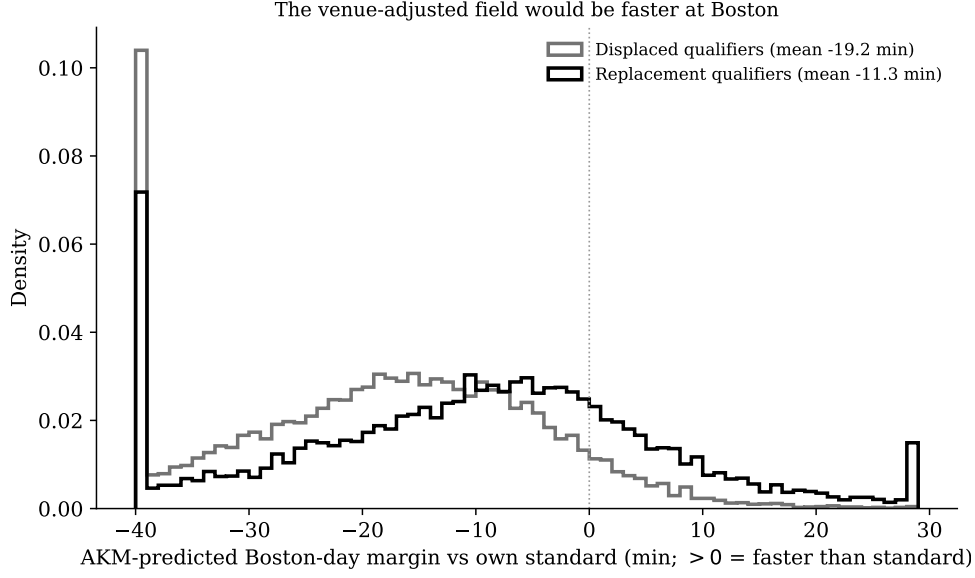


Figure 12: Predicted Boston-day performance of displaced vs. replacement qualifiers (AKM runner effects; margins relative to each runner’s own standard).

through the runners who use course c strategically, so the contamination scales with the course’s *strategic share* s_c —the share of its finishers whose qualification depends on the course—which is directly observable. To first order,

$$\psi_c^{\text{naive}} = \delta_c + s_c b_c + \text{noise}, \quad (3)$$

where b_c is the average behavioral residual of the course’s strategic users. The median course has a strategic share of about 1.0% of finishers: for the overwhelming majority of venues the naive index *is* the physical index, and contamination concentrates exactly where qualification is the product being sold—the purpose-built qualifying venues whose strategic shares reach an order of magnitude higher. The diagnostics below estimate b_c (order and peaking asymmetries), measure s_c , and combine them via equation (3); composition controls provide independent corrections, and the index suite bounds what remains.

8.1 Order Effects, Edges, and Peaking

Additive separability holds on average but not edge by edge, and the violations are informative behavior, not noise. Following the movers-design logic of Card, Heining and Kline (2013), define for each ordered same-runner pair $r \rightarrow s$ the residual $\eta_{r \rightarrow s} = (\ln T_s - \ln T_r) - (\hat{\psi}_s - \hat{\psi}_r)$: the part of the performance change not explained by the additive course difference. Its mean is near zero—the additive model fits well on average—but it is systematically positive *into* Boston: runners are about +7.9 minutes slower at Boston than the additive model predicts (they peaked at the qualifier), versus −2.1 minutes when Boston is the *origin*—an asymmetry of +10.1 minutes (Appendix Figure A1a, Appendix Table A6; the asymmetry survives to two-way clustering by runner and course). A fixed physical feature would be symmetric; this is behavior.⁴ The effect is Boston-specific—it does not appear into lottery-entry placebo destinations (Appendix Table A5 reports the destination-by-destination comparison)—and reverses with direction (Appendix Figure A1b), and the peaking interpretation has a direct signature: runners who *qualified* at the origin race are +12.5 minutes slower at their next race than the additive model predicts—the flattering performance attaches to the race where the qualifying campaign peaked.

Directional mover asymmetry is a general diagnostic, not a marathon curiosity: motivated cohorts sorting into tested grades and elective patients traveling to flagship hospitals generate the same signature—the same pair of venues implies different gaps depending on order—and the same ordered-pair residuals detect it.

8.2 Strategic Shares and the Behavior-Adjusted Index

The strategic share is the proportionality constant in equation (3), and measuring it shows where contamination can and cannot live. At St. George, a steeply downhill course, course-

⁴Three numbers measure this Boston-state phenomenon on different samples: the Boston-retest regression constant (Appendix A), the mean course-level additivity residual (Appendix C), and the ordered-pair coefficient +7.9 here. They differ because the first two pool directions and partial out different controls; the ordered-pair estimate is the directional measure the behavior adjustment uses.

dependent qualifiers are about 5.6% of finishers; the median among courses with a measured peaking residual is 1.0%; only the purpose-built qualifying products reach contamination-relevant shares (Appendix Table A7). Because strategic users are a minority almost everywhere, they barely move the course effect: re-estimating the model after excluding near-threshold observations (the “clean” index) leaves the benchmark downhill courses essentially unchanged. A steeply downhill course is genuinely fast for everyone who runs it.

The behavior-adjusted index implements equation (3), netting out the course-level peaking contamination by scaling the origin peaking residual η_c by the strategic share:

$$\hat{\delta}_c^{\text{adj}} = \hat{\delta}_c + s_c \eta_c. \quad (4)$$

The adjustment is deliberately modest and agrees with the clean re-estimate—St. George moves from -6.4 to -5.7 minutes relative to Boston, and the typical course barely moves. Only a course’s strategic users additionally peak, so only their share of the field is netted out. Applying the raw retest residual to the whole course instead would be a selection error: the residual is estimated from the self-selected minority who qualified and went to Boston, and attributing their behavior to every finisher would absurdly flip genuinely fast courses to slower-than-Boston. Section 7 used the behavior-adjusted index as the headline screen, with the full suite as the robustness band.

Figure 13 plots the per-course correction against the strategic share, summarizing the section’s proportionality result. The correction’s magnitude is essentially monotone in s_c (Spearman 0.94 among courses with a measured peaking residual), every course whose correction exceeds one minute has a strategic share above about 8%, and 99% of US road finishers run courses where the total correction is under one minute. For the overwhelming majority of venues the naive index is the physical index, and the exceptions are the purpose-built qualifying products themselves—which strengthens, rather than hedges, the paper’s thesis.

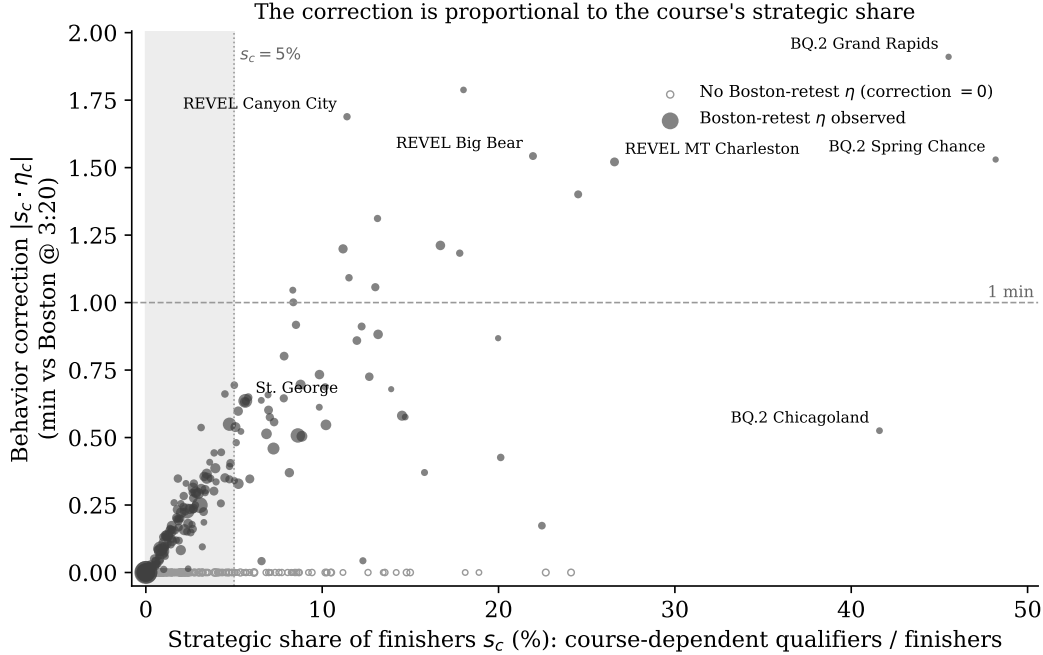


Figure 13: Behavior correction is proportional to the strategic share. Each point is a US road course: the horizontal axis is the strategic share s_c (course-dependent qualifiers as a share of finishers), the vertical axis the behavior correction in minutes at 3:20 pace, and marker size is proportional to finisher count. Courses without a measured peaking residual receive no correction by construction (open circles); shading marks $s_c < 5\%$; purpose-built qualifying venues are labeled.

8.3 Composition Controls: Routine Attempts and Reverse Pairs

A natural alternative to correcting the pooled index is to split the sample and estimate the model separately for targeting and non-targeting attempts. Targeting, however, is a runner-*race* state, not a runner type: the same person targets in some attempts and not others, so the pair, not the runner, is the right unit of a split. The routine-attempt estimator implements that split. Define an attempt as *routine* ex ante if the runner’s geocoded home is within 100 miles of the course or the runner has run that course in a prior year: locals and repeaters run Chicago because it is their race, not as a once-peaked goal. Estimating course effects from routine–routine same-runner pairs only (a within-runner pair estimator on 1.9 million pairs) gives a routine-only index whose correlation with the naive index is 0.96 across 733 courses. Each course’s *visitor premium*—naive minus routine-only,

after removing the common Boston-numeraire level that the invariance result (Appendix A) makes irrelevant—measures how much of its apparent speed is visitor composition. The premium concentrates where peaking predicts: purpose-built goal venues are flattered by visitor composition (REVEL Big Bear by about -6.8 minutes), ordinary majors are not, and the interquartile range across courses is -2.7 to +1.5 minutes (Appendix Figure A2, Appendix Table A8). The contamination is real, signed as predicted, nameable course by course—and bounded. The split fails only where targeting is the product itself (who runs a last-chance qualifier casually?), which is why the suite below pairs it with corrections that need no non-targeting subsample.

A second composition control uses direction. Averaging the $A \rightarrow B$ and $B \rightarrow A$ directions of same-pair gaps differences out order and peaking components that are direction-specific, identifying $\delta_B - \delta_A$ under the assumption that order effects are symmetric in expectation. This *reverse-pair* estimator (the constructive use of Appendix Figure A1b) joins the suite in Appendix Table C7.

8.4 The Corrected-Index Suite

Table 2 collects the main indices, what each neutralizes, and the fixed-field displacement each implies; the policy accounting of Section 7 reported its headline against this band. Three further variants—the clean re-estimate excluding the threshold region, the reverse-pair symmetric estimator, and a conservative shrinkage that zeroes statistically imprecise courses—appear in Appendix Table C7. Across all 7 indices, built on one consistent sample, displacement ranges from 10.5% to 14.9%, so the policy conclusion does not hinge on a unique “true” index. One limitation qualifies the band’s interpretation: the observable-predicted index is fit to contaminated fixed effects, and peaking correlates with fast terrain, so it is not a fully behavior-free benchmark either—the band, not any single index, is the answer.

Table 2: The course-speed index suite: construction, what each neutralizes, and the implied fixed-field displacement (US 2018–19 qualifiers, field size 85,466). Appendix Table C7 reports the full 7-index suite.

Index	Neutralizes	Displaced (%)
Behavior-adjusted (headline), eq. (4)	Course-level peaking of strategic users	13.9
Naive AKM ψ (starting point)	—	14.6
Routine-attempt	Visitor/goal-race composition	12.1
Observable-based (terrain)	All non-physical variation (and some physical)	10.5

Notes: The four main indices; the remaining three (clean re-estimate, reverse-pair symmetric, conservative shrinkage) appear with these in Appendix Table C7 and in Figure 11, Panel B. Displacement is the one-sided share of the fixed field reallocated when candidates are ranked by the index-adjusted margin.

8.5 Conservative and observable-based variants

Two implementable variants discipline estimation error. A confidence-adjusted handicap shrinks each course’s adjustment toward zero by k standard errors, zeroing courses statistically indistinguishable from Boston; at $k = 1.645$ displacement falls to 11.9%. An observable-based rule uses predicted speed from terrain, surface, and climate—applicable to brand new courses with no race history—and yields 10.5%; it under-adjusts the purpose-built fast-loop courses whose speed does not come from elevation, which is also why it is not a behavior-free benchmark. A practical implementation uses the observable formula for new courses and the estimated index for established ones, updating annually.

8.6 The case against the net-drop proxy

The BAA’s 2027 rule penalizes net elevation drop alone, which explains $R^2 = 0.05$ of cross-course speed variation. It therefore misses on both sides—the fast-flat purpose-built products

escape unpenalized while some large-net-drop courses with heavy climbing are overcorrected (Section 6.4)—and any single-proxy policy leaves most venue heterogeneity untouched, inviting re-sorting toward the unpenalized fast venues.

8.7 Rule classes and the next round

The three available rule classes create different next-round margins. A *cardinal handicap* (this paper’s index) equalizes measured venue speed but inherits estimation error and requires re-estimation as courses change. An *ordinal within-venue rule*—qualify the top $X\%$ of each race’s field, the pay-for-percentile logic of Barlevy and Neal (2012)—is robust to any common venue effect by construction (ranking within a race cancels whatever is common to everyone in it), but creates a new margin: shopping for weak fields, which favors small races, where fields are thin and manipulable. A *proxy rule* (net drop) is transparent but, as above, barely correlated with the object it targets. The choice among them is a choice among induced behaviors, not just among approximations to δ_c . Corrections must also choose which components to neutralize—venue physics yes, effort no—and any published correction becomes the next target.

Reweight near-BQ runners to the inframarginal course distribution—the no-incentive counterfactual—removes about half of the near-BQ fast-course excess and lowers the misallocation rate among apparent qualifiers by roughly five percentage points in 2018–19; equilibrium exit of marginal demand would additionally discourage fast-course entry, so the fixed-choice numbers above are a lower bound on the policy’s full effect. The bound’s assumption is that inframarginal runners’ venue distribution is not itself fast-tilted for non-BQ reasons. Finally, any published handicap opens the next round: runners can chase courses with imprecise or stale handicaps, and suppliers can design to the formula. How often the index is re-estimated, whether handicaps apply *ex ante* or *ex post*, and how new courses are scored will determine how much of the screening distortion survives. A static handicap reduces today’s misallocation; it does not by itself close the loop.

9 Conclusion

A common qualifying threshold applied across heterogeneous, manipulable evaluation environments produces two linked distortions, and the observed venue value-added a naive fixed-effects index reports is a composite of physical venue speed and the behavior the threshold induces. This paper builds the index, tests its assumptions in the open, documents the market the threshold created, prices the misallocation, and corrects the measure for the behavior it absorbs.

The screening distortion is large: more than one in five runners on the Boston start line (23.2% in the 2026 cohort) qualified only by virtue of the course they chose, and about 13.9% of recent Boston qualifiers would be displaced under a venue-adjusted standard—robustly 10.5–14.9% across the index suite, roughly an order of magnitude above what estimation noise alone can generate—and replaced by runners predicted to be about 7.9 minutes faster at Boston itself. The market distortion is two-sided: near-threshold runners sort to fast venues, with the response jumping at announced tightenings; and the supply of fast venues expanded 2.3-fold, with purpose-built qualifying products entering until the supply channel contributed 4.5 percentage points of the misallocation rate—of the same order as candidate sorting under every counterfactual baseline. The screening rule does not merely rank runners imperfectly; by the end of the sample period it had reshaped the market in which qualifying performances are produced.

These distortions cannot be repaired by targeting a few extreme courses: net elevation drop, the BAA’s chosen proxy, explains almost none of the cross-course variation in speed. Index-based venue adjustment is coherent and implementable, ordinal within-venue rules are robust to venue effects at the cost of a new field-shopping margin, and every published correction opens a next round.

For measurement practice, the marathon market is a laboratory in which the full Goodhart loop around a fixed-effects performance measure is visible end to end, with ground truth on venue physics and a retest at the prize venue. The tools generalize: when a

fixed-effect measure becomes the prize, exogenous mobility fails by design, predictably, and locally—signed toward favorable venues, concentrated near the threshold, intensifying with each tightening—which makes it detectable through threshold-distance interactions, moving-threshold and placebo-threshold tests, and directional mover asymmetries; strategic shares scale the correction; and audits of candidate gaming alone miss the supply side, which here grew to first order within a decade.

One caveat raises the credibility of the rest: detection in this setting is powered by a threshold that is public, sharp, and moves on announced dates. Where operative thresholds are murkier—teacher evaluation, hospital report cards—the toolkit’s power is correspondingly an upper bound. What carries over is the logic: measure the components, exploit the institution’s own discontinuities, and audit both sides of the market.

The broader lesson is that common thresholds are not neutral when performance is produced in environments candidates choose and suppliers build. In admissions, accountability, certification, and ratings, the same loop operates with the environments harder to see; here, every link of the loop—rule, candidates, venues, and measure—is observed.

References

- Abowd, John M., Francis Kramarz, and David N. Margolis.** 1999. “High Wage Workers and High Wage Firms.” *Econometrica*, 67(2): 251–333.
- Allen, Eric J., Patricia M. Dechow, Devin G. Pope, and George Wu.** 2017. “Reference-Dependent Preferences: Evidence from Marathon Runners.” *Management Science*, 63(6): 1657–1672.
- Andrews, Martyn J., Leonard Gill, Thorsten Schank, and Richard Upward.** 2008. “High Wage Workers and Low Wage Firms: Negative Assortative Matching or Limited Mobility Bias?” *Journal of the Royal Statistical Society Series A*, 171(3): 673–697.
- Bagues, Manuel F., Mauro Sylos-Labini, and Natalia Zinovyeva.** 2008. “Differential Grading Standards and University Funding: Evidence from Italy.” *CESifo Economic Studies*, 54(2): 149–176.
- Baker, George P.** 1992. “Incentive Contracts and Performance Measurement.” *Journal of Political Economy*, 100(3): 598–614.
- Barlevy, Gadi, and Derek Neal.** 2012. “Pay for Percentile.” *American Economic Review*, 102(5): 1805–1831.
- Bolton, Patrick, Xavier Freixas, and Joel Shapiro.** 2012. “The Credit Ratings Game.” *Journal of Finance*, 67(1): 85–112.
- Bonhomme, Stéphane, Thibaut Lamadon, and Elena Manresa.** 2019. “A Distributional Framework for Matched Employer Employee Data.” *Econometrica*, 87(3): 699–739.
- Boston Athletic Association.** 2025a. *2027 Boston Marathon Qualifying Standards and Course Adjustment Policy*. BAA. <https://www.baa.org/races/boston-marathon/qualify>.
- Boston Athletic Association.** 2025b. “Field of Qualifiers Notified of Acceptance into the 130th Boston Marathon presented by Bank of America.” *B.A.A. press release*, <https://www.baa.org/news/130thqualifiers/>.
- Campbell, Donald T.** 1979. “Assessing the Impact of Planned Social Change.” *Evaluation and Program Planning*, 2(1): 67–90.
- Card, David, Jörg Heining, and Patrick Kline.** 2013. “Workplace Heterogeneity and the Rise of West German Wage Inequality.” *Quarterly Journal of Economics*, 128(3): 967–1015.
- Chetty, Raj, John N. Friedman, and Jonah E. Rockoff.** 2014a. “Measuring the Impacts of Teachers I: Evaluating Bias in Teacher Value-Added Estimates.” *American Economic Review*, 104(9): 2593–2632.

- Chetty, Raj, John N. Friedman, and Jonah E. Rockoff.** 2014b. “Measuring the Impacts of Teachers II: Teacher Value-Added and Student Outcomes in Adulthood.” *American Economic Review*, 104(9): 2633–2679.
- Correia, Sergio.** 2017. “Linear Models with High-Dimensional Fixed Effects: An Efficient and Feasible Estimator.” *Working Paper*. <http://scorreia.com/research/hdfe.pdf>.
- Dee, Thomas S., Will Dobbie, Brian A. Jacob, and Jonah Rockoff.** 2019. “The Causes and Consequences of Test Score Manipulation: Evidence from the New York Regents Examinations.” *American Economic Journal: Applied Economics*, 11(3): 382–423.
- Diamond, Rebecca, and Petra Persson.** 2016. “The Long-term Consequences of Teacher Discretion in Grading of High-Stakes Tests.” National Bureau of Economic Research Working Paper 22207.
- Dranove, David, Daniel Kessler, Mark McClellan, and Mark Satterthwaite.** 2003. “Is More Information Better? The Effects of “Report Cards” on Health Care Providers.” *Journal of Political Economy*, 111(3): 555–588.
- Goodhart, Charles A.E.** 1984. “Problems of Monetary Management: The U.K. Experience.” *Monetary Theory and Practice*, 91–121.
- Jacob, Brian A.** 2005. “Accountability, Incentives and Behavior: The Impact of High-Stakes Testing in the Chicago Public Schools.” *Journal of Public Economics*, 89(5–6): 761–796.
- Kleven, Henrik J., and Mazhar Waseem.** 2013. “Using Notches to Uncover Optimization Frictions and Structural Elasticities: Theory and Evidence from Pakistan.” *The Quarterly Journal of Economics*, 128(2): 669–723.
- Kline, Patrick, Raffaele Saggio, and Mikkel Sølvesten.** 2020. “Leave-Out Estimation of Variance Components.” *Econometrica*, 88(5): 1859–1898.
- Lazear, Edward P.** 2006. “Speeding, Tax Fraud, and Teaching to the Test.” *Quarterly Journal of Economics*, 121(3): 1029–1061.
- MarathonGuide.com.** 2024. “MarathonGuide.com: Race Results Database.” <https://www.marathonguide.com>.
- Morris, Carl N.** 1983. “Parametric Empirical Bayes Inference: Theory and Applications.” *Journal of the American Statistical Association*, 78(381): 47–55.
- Neal, Derek, and Diane Whitmore Schanzenbach.** 2010. “Left Behind by Design: Proficiency Counts and Test-Based Accountability.” *Review of Economics and Statistics*, 92(2): 263–283.
- Rothstein, Jesse.** 2010. “Teacher Quality in Educational Production: Tracking, Decay, and Student Achievement.” *Quarterly Journal of Economics*, 125(1): 175–214.

- Sabot, Richard, and John Wakeman-Linn.** 1991. “Grade Inflation and Course Choice.” *Journal of Economic Perspectives*, 5(1): 159–170.
- Saez, Emmanuel.** 2010. “Do Taxpayers Bunch at Kink Points?” *American Economic Journal: Economic Policy*, 2(3): 180–212.

Appendix A Measurement Details: AKM Specification, Shrinkage, and Course Drivers

This appendix gives the AKM specification and Empirical Bayes shrinkage behind the course-speed index of Section 3, an algebraic invariance result for the displacement accounting, the within-runner additive-separability validation behind Section 4.1, the observable predictors of course speed, and the contamination-diagnostics exhibits behind Section 8.

A.1 AKM Estimator

I estimate the following specification:

$$\ln T_{irt} = \alpha_i + \delta_r + \gamma_t + f(\text{age}_{it}, \text{male}_i) + g(W_{rt}) + \varepsilon_{irt} \quad (5)$$

where T_{irt} is the finish time (in seconds) of runner i in race r in year t , α_i is a runner fixed effect capturing persistent baseline performance, δ_r is a race fixed effect capturing the persistent speed of course r (including its typical climate), and γ_t is a year fixed effect capturing common trends across all races. The function $f(\cdot)$ is a quartic polynomial in age, centered at 40, fully interacted with gender. The term $g(W_{rt})$ is a quadratic in demeaned race-day temperature, $g(W_{rt}) = \beta_1 \widetilde{W}_{rt} + \beta_2 \widetilde{W}_{rt}^2$, where $\widetilde{W}_{rt}$ is the deviation of race-day morning temperature from the marathon’s historical mean. Because mean temperature is absorbed by the course fixed effect δ_r , the weather controls capture only transitory race-year temperature shocks. (Humidity, wind, and precipitation add little explanatory power *on average* and are omitted from the model; they can matter at specific exposed courses, which is visible in the within-course stability diagnostic of Appendix D.)

To reduce false links in the entity resolution procedure, I define runner identities separately within broad geographic regions (U.S., Canada, UK/Ireland, Australia/New Zealand, and Other), so the same physical runner competing in both the U.S. and UK is assigned

separate fixed effects in each region. The model is estimated using `reghdfe` (Correia, 2017), with standard errors clustered by regional runner.

The key assumption is additive separability: after accounting for runner ability and observables, courses shift log finish times by a common amount. This is a plausible approximation for road marathons, where features such as elevation profile, terrain, and typical weather conditions tend to affect all runners proportionally in time rather than creating highly runner-specific match effects. Appendix D reports two diagnostic exercises in the spirit of Card, Heining and Kline (2013): within-course year-to-year stability of course effects, and a nonseparability test. Additive separability (Section 4.1 in the main text; regression detail in Section A.4; $\hat{\beta} \approx 0.851$ for Boston-destination and 0.933 for general cross-race pairs) supplies the mover-style cross-venue validation that is typically the third CHK diagnostic. Taken together, these exercises suggest that the additive structure is a useful approximation, though not an exact description of every runner-course combination.

The overall R^2 of equation (5) is 0.816. Table A1 reports a progressive specification that demonstrates the contribution of each set of fixed effects. Demographics alone explain 11% of log-time variation. Adding runner fixed effects raises R^2 to 0.83, reflecting the dominance of persistent performance differences. Race fixed effects contribute an additional 2 percentage points, modest in levels but substantively important, as these capture the course speed variation that is the object of interest. Year fixed effects and weather controls add a small further increment. The weather controls are significant: a 10°F warmer-than-average race day slows finish times by 1.8%, or about 3.6 minutes at a 3:20 pace.

Table A1: Progressive specification: sources of variation in log finish time

	(1)	(2)	(3)	(4)
	Demographics	+ Runner FE	+ Race FE	+ Year FE + Weather
R^2	0.109	0.829	0.846	0.849
Adj. R^2	0.109	0.770	0.793	0.796
RMSE	0.198	0.100	0.095	0.095
Observations	7,725,138	7,723,435	7,723,418	7,723,418

Notes: Dependent variable is log finish time. Demographics are a quartic in age interacted with gender. Runner, race, and year fixed effects are added progressively. Column (4) adds year fixed effects and a quadratic in demeaned race-day temperature. Standard errors clustered by regional runner identifier.

A.2 From Fixed Effects to a Course Speed Index

The course speed index is constructed from the estimated race fixed effects $\hat{\delta}_r$, which capture persistent differences in venue advantage after netting out runner ability, age-gender profiles, and race-year shocks. Because the model is estimated separately within each geographic region, these fixed effects are identified only up to a region-specific constant. I normalize the U.S. estimates by setting $\hat{\delta}_{\text{Boston}} = 0$ and calibrate the remaining regions using approximately 24,000 trusted bridge runners who appear in both the U.S. and another region. For each non-U.S. region, I use the median difference in runner fixed effects among bridge runners to recover a calibration constant that places all course effects on the Boston-centered U.S. scale.

I then apply Empirical Bayes shrinkage to the estimated course effects to reduce noise from sparsely connected courses. Intuitively, the procedure treats each estimated $\hat{\delta}_r$ as a noisy signal of an underlying permanent course effect and pulls imprecisely estimated values toward the overall mean, with stronger shrinkage for courses estimated with greater variance

(Morris, 1983; Andrews et al., 2008). Letting $\widehat{V}_r$ denote the estimated sampling variance of $\hat{\delta}_r$ and $\hat{\tau}^2$ the cross-course variance of the underlying effects, the shrinkage factor is

$$w_r = \frac{\hat{\tau}^2}{\hat{\tau}^2 + \widehat{V}_r},$$

so the shrunk estimate is

$$\hat{\delta}_r^{shr} = w_r \hat{\delta}_r.$$

Courses with little information are therefore pulled more strongly toward zero, while well-measured courses are left largely unchanged. I retain courses with at least 50 AKM observations after shrinkage; the final index covers 1057 courses.

Because the model is estimated in logs, the course effect $\hat{\delta}_r^{shr}$ is a proportional measure: a course with $\hat{\delta}_r^{shr} = -0.05$ is approximately 5% faster than Boston for all runners, regardless of pace. This proportional property means a given percentage advantage translates to more minutes for slower runners. For a runner on pace to finish in 3:00, a 5% course advantage is worth 9 minutes; for a 3:20 runner, 10 minutes; for a 4:00 runner, 12 minutes.

Throughout the paper, I report course effects in percentage terms and provide minute equivalents at two reference paces: 3:00, the 2025 M18–34 qualifying standard, and 3:20, which is the pooled mean qualifier finish time (median 3:18:42) across 929,907 qualifier-races in the sample. I define a “fast” course as one that is at least 2% faster than Boston—equivalent to about 3.6 minutes at a 3:00 pace or 4 minutes at a 3:20 pace—which captures 237 courses (26% of the index). I define a “very fast” course as one at least 4% faster (82 courses, 9%). I define the BQ handicap as the percentage by which the qualifying standard should be changed to equalize course difficulty; because it is proportional, a single handicap applies to all age and gender groups.

A.3 An Invariance Result

One algebraic fact disciplines the displacement accounting. The fixed-field counterfactual in Section 7 ranks all candidates by venue-adjusted time $T_i e^{-\hat{\delta}_{r(i)}}$ and admits a fixed number. Adding any constant k to every course effect rescales every adjusted time by e^{-k} and leaves the ranking—hence the admitted set, hence the displacement share—unchanged. The ranking invariance has three consequences. First, the choice of numeraire cannot affect the displacement headline: whether “Boston = 0” means Boston in race-day state or Boston in qualifying-attempt state is irrelevant. Second, any economy-wide component of effort or peaking absorbed into the index is irrelevant. Third, *only cross-course differential contamination matters*—which is precisely what the diagnostics of Section 8 measure and what the corrected-index suite bounds. The misallocation *rate* against the posted standard is not invariant to k ; that series is therefore always presented with its Boston-state numeraire stated, and the cross-index band serves as its robustness check.

A.4 Validation: Cross-Race Additive Separability and the Boston Retest

The estimated course effects are useful only if they capture real, additively separable differences in venue speed. The cleanest test is whether the index is predictive of subsequent Boston race performance. The regression model below assesses this prediction. I then generalize to the full set of cross-race pairs as a broader check.

The core specification, applied throughout this subsection, is:

$$\ln T_{i,\text{dest}} - \ln T_{i,\text{orig}} = \beta \cdot (\hat{\delta}_{\text{dest}} - \hat{\delta}_{\text{orig}}) + \mathbf{X}_i' \boldsymbol{\phi} + \varepsilon_i \quad (6)$$

where $\hat{\delta}_{\text{dest}} - \hat{\delta}_{\text{orig}}$ is the difference in estimated course effects between the destination and origin races and $\mathbf{X}_i$ collects a quadratic in age and gender. By differencing finish times within runner, this specification absorbs time-invariant runner ability. Perfect additive separability

implies $\beta = 1$: moving to a course that is 1 log point faster should improve finish time by 1 log point on average. Standard errors are clustered by runner.

A.4.1 Boston as a Standardized Retest

Because Boston is the normalization point of the course index, $\hat{\delta}_{\text{Boston}} = 0$ by construction. I construct the retest sample by taking every runner-year in which the runner ran Boston and pairing it with each of that runner’s marathons in the 18 months before Boston at a different course, yielding 690,058 pairs. On average, runners are slower at Boston than at their prior race, and faster qualifying courses predict larger Boston slowdowns (Figure 2, Panel A).

Table A2, column 2, reports estimates of equation (6) on the Boston-destination subsample. The estimated additive separability coefficient is $\hat{\beta} = 0.851$: for each 1% increase in a prior marathon’s speed, a runner is predicted to run Boston about 1% slower. On a raw basis, runners are on average slower at Boston than at their prior race; most of this is explained by course-FE additive separability, leaving a positive constant that captures residual Boston-specific factors such as fatigue, travel, race-day conditions, and the order/peaking behavior measured directly in Section 8.1. (Figure 2 in the main text plots both retest panels.)

A.4.2 General Cross-Race Additive Separability

To show that additive separability is not specific to Boston, I generalize the pair construction to all destinations. For every runner-race observation in the dataset, I pair it with each prior marathon by the same runner in the 18 months before at a different course.⁵ This yields 10,934,525 pairs. Figure 2, Panel B, shows the binscatter; Table A2, column 1, reports $\hat{\beta} = 0.933$. Column 4 estimates an interacted model that allows the slope to differ between Boston and non-Boston destinations; both the Boston-destination ($\hat{\beta} = 0.851$) and non-

⁵The same-course restriction excludes pairs like Boston 2019 / Boston 2020, which would contribute a degenerate zero FE-difference observation; the runner still enters the analysis through cross-course pairs (e.g., Boston 2019 / NYC 2018 and Boston 2020 / NYC 2018).

Boston-destination ($\hat{\beta} = 0.934$) coefficients closely approach the 1.0 benchmark of perfect additive separability.

Table A2: Cross-race additive separability of course fixed effects

	(1)	(2)	(3)	(4)
	All pairs	Boston dest.	Non-Boston	Interacted
FE difference	0.933*** (0.002)	0.851*** (0.009)	0.934*** (0.002)	
FE diff $\times$ Boston				0.852*** (0.009)
FE diff $\times$ non-Boston				0.934*** (0.002)
Boston-dest indicator				0.042*** (0.000)
Age controls	✓	✓	✓	✓
N	10,934,525	690,058	10,244,467	10,934,525
R^2	0.156	0.070	0.161	0.162

Notes: Dependent variable is the log time gap, $\ln T_{\text{dest}} - \ln T_{\text{origin}}$. “FE difference” is the course fixed effect at the destination minus the origin. Each runner’s destination race is paired with every prior marathon at a different course within 18 months; pair-weighted WLS so prolific runners do not dominate; age and gender controls. Column (4) interacts the FE difference with a Boston-destination indicator. Standard errors clustered by runner in parentheses. *** $p < 0.001$.

Table A3: Additive-separability slope by pair stratum: name rarity and entity-resolution link provenance

Pair stratum	All destinations			Boston destination		
	$\hat{\beta}$	(SE)	Pairs	$\hat{\beta}$	(SE)	Pairs
All pairs	0.933	(0.001)	10,934,525	0.851	(0.004)	690,058
Rare names (≤ 2 people per name)	0.895	(0.001)	6,963,864	0.774	(0.004)	482,583
Medium (3–10 people)	1.006	(0.002)	2,939,352	0.961	(0.009)	152,046
Common (> 10 people)	1.162	(0.003)	1,031,309	1.357	(0.019)	55,429
Excluding attached yob-less links	0.936	(0.001)	5,272,574	0.857	(0.005)	441,583
Excluding residence-merge links	0.915	(0.001)	4,861,866	0.810	(0.005)	387,801
Medium, all low-confidence links excluded	0.981	(0.003)	710,472	0.928	(0.014)	62,513
Rare, all low-confidence links excluded	0.876	(0.001)	2,098,419	0.741	(0.006)	208,080

Notes: WLS estimates of equation (6) within pair strata (pair-weighted, age and gender controls). Tiers classify the runner’s (first name, last name, gender) cell by the number of distinct people carrying it, measured as distinct birth-year clusters. “Attached yob-less links” are runner ids that mix rows with and without a birth interval (the entity resolution’s NA-attach step); “residence-merge links” are ids consolidated by the modal-residence re-merge. Slopes above one among common names indicate occasional false merges—a spurious mover pairing a faster person with a faster course—while slopes into Boston sitting below slopes elsewhere indicate origin-race peaking (Section 8). Excluding the entity-resolution merge steps leaves the slopes essentially unchanged, so the deviations are not artifacts of the linkage design.

A.5 What Drives Course Speed?

Course fixed effects capture everything that makes a marathon systematically faster or slower after accounting for runner ability and demographics just by comparisons of runners as they switch courses, but that also makes it a “black box” and difficult to understand why course speeds are different. To ask whether the estimated course effects line up with observable geographic and natural features—and in particular whether “downhill” is enough to explain the relevant variation—I collect detailed course profiles for the 529 courses with at least 500

observations and available profile data, covering elevation (gross gain, net drop, maximum elevation), course geometry, surface, climate, exposure, setting, and logistics. I then estimate cross-sectional regressions of the form:

$$\hat{\Delta}_r = \mathbf{X}_r' \boldsymbol{\beta} + u_r \quad (7)$$

where $\hat{\Delta}_r$ is the estimated course effect in minutes and $\mathbf{X}_r$ is a vector of observable course characteristics.

Table A4 reports three specifications of equation (7). Net elevation drop alone explains only about 5% of the variation ($R^2 = 0.05$). Gross elevation gain is far more important ($R^2 = 0.73$). A kitchen-sink model with altitude, course type, surface, and climate raises R^2 to 0.83—but 15% remains unexplained, reflecting features such as crowding, wind exposure, course layout, field selection, multi loop courses, or runner support that are captured by the AKM effect but difficult to summarize with simple observables.

Table A4: Observable predictors of course speed: net drop misses most of the variation

	(1)	(2)	(3)
	Net drop only	Gross gain only	Kitchen sink
Net elev. drop (1000 ft)	-0.03*** (0.01)		-0.02*** (0.00)
Gross elev. gain (1000 ft)		0.06*** (0.00)	0.04*** (0.01)
Max elevation (1000 ft)			0.01*** (0.00)
Trail surface			0.08*** (0.02)
Avg. race temp (°F)			0.00*** (0.00)
<i>N</i>	529	529	529
<i>R</i> ²	0.052	0.762	0.850

Notes: Dependent variable is the log course speed effect (EB-shrunk AKM estimate, Boston = 0). Positive values indicate slower courses. Sample is the 529 courses with at least 500 AKM observations and complete elevation, surface, and climate data. Robust (HC1) standard errors in parentheses. Column (3) additionally includes course type, coastal, rural, prize money, and pacer program indicators. A coefficient of 0.06 on gross gain means each 1,000 feet of cumulative climbing is associated with about 6% slower finish times, or approximately 12 minutes at a 3:20 pace. Values from `course_profile_regressions.csv`; the temperature coefficient is positive and precise but rounds to 0.00 at this display precision. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

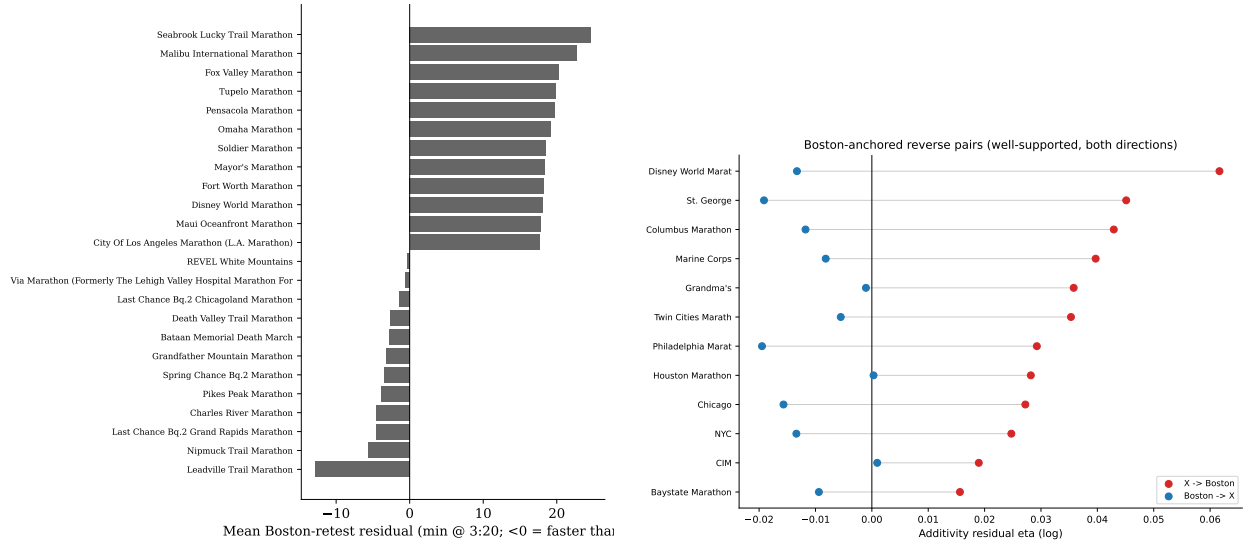
A.6 Contamination Diagnostics: Additional Exhibits

This subsection collects the exhibits behind Section 8: the destination-numeraire retest slopes, the ordered edge/peaking diagnostics, the strategic-share and behavior-adjusted-

index detail, and the routine-attempt visitor premia.

Table A5: Boston retest by destination numeraire: same-runner pairs into each destination; slope of the realized log-time change on the predicted course difference. SEs clustered by runner.

Destination race	Slope $\hat{\beta}$	SE	Pairs
Boston	0.851	(0.009)	690,058
Chicago	0.988	(0.035)	11,852
New York City	0.998	(0.038)	2,438
Marine Corps	0.826	(0.039)	62,607
Philadelphia	0.824	(0.046)	19,941
All destinations	0.933	(0.002)	10,934,525



(a) Boston-retest residual by US origin course (b) Direction matters (reverse-pair asymmetry)

Figure A1: Runners peak at the qualifier, not at Boston: the flattering performance attaches to the race run before Boston, and the residual reverses with direction.

Table A6: Ordered edge/order decomposition of the additivity residual η (minutes at 3:20 pace). Columns add controls progressively; SEs clustered by runner (two-way runner $\times$ course SEs reported in the underlying output).

	(1) Order	(2) +Near-BQ	(3) +Goal/qual.
Destination is Boston	7.930	7.396	2.907
Origin is Boston	-2.134	-3.325	-6.371
Near BQ at origin		9.869	5.419
Dest. Boston $\times$ near BQ at origin		-6.062	-1.591
Qualified at origin			12.490
Dest. Boston $\times$ qualified at origin			-3.245
Major-city destination			0.296
Temperature deviation difference ($^{\circ}$ F)	0.344	0.340	0.333

Table A7: Strategic share and the behavior-adjusted index for the fastest courses. η is the Boston-retest peaking residual; “Adjusted” is equation (4).

Course	Naive	Strategic share (%)		Peaking η	Adjusted
	(min)	Finishers	Qualifiers	(min)	(min)
REVEL MT Charleston	-17.1	26.6	77.6	6.3	-15.6
Spring Chance BQ.2 Marathon	-16.6	48.2	90.0	-3.4	-18.2
Tunnel Lite Marathon	-15.6	22.5	63.5	0.8	-15.4
Last Chance BQ.2 Chicagoland Marathon	-15.3	41.6	83.9	-1.4	-15.8
REVEL Big Bear and Half Marathon	-15.1	22.0	66.5	7.7	-13.6
Last Chance BQ.2 Grand Rapids Marathon	-14.5	45.5	87.8	-4.5	-16.4
Light At The End Of The Tunnel Marathon	-12.8	24.5	71.3	6.2	-11.4
MT. Nebo Marathon	-11.4	9.8	52.1	6.7	-10.8
Hudson Mohawk Marathon	-11.2	17.8	56.0	7.1	-10.0
Carmel Marathon	-11.2	12.0	52.0	7.7	-10.3
Loco Marathon	-10.0	14.7	58.7	4.2	-9.4
Tunnel Vision Marathon	-9.9	20.1	60.1	2.2	-9.5
REVEL Canyon City Marathon	-9.6	11.4	51.9	16.1	-7.9
Bay State Marathon	-9.6	18.0	55.9	10.6	-7.8

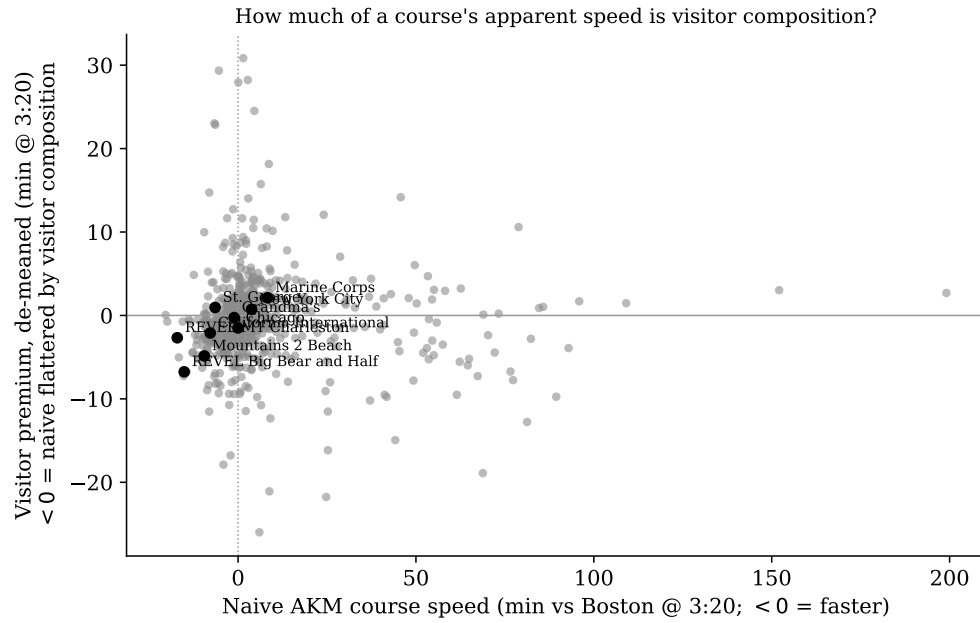


Figure A2: De-meaned visitor premium by course: negative values mean the naive index is flattered by visitor composition. Purpose-built goal venues stand out; ordinary local races sit near zero.

Table A8: Routine-only course speeds and visitor premia, largest absolute de-meanned premia among well-measured courses.

Course	Naive (min)	Routine-only (min)	Visitor premium (min)	Routine share (%)	Routine pairs
Bataan Memorial Death March	78.9	56.5	10.6	27.0	4,420
Sarasota Marathon The Grouper Run	0.7	3.6	-9.1	53.9	1,551
Loonies Midnight Marathon	2.2	-12.6	9.0	30.2	1,102
Lakeshore Marathon	-1.8	0.0	-8.0	51.5	2,899
Ocala Marathon	-0.8	0.7	-7.8	50.2	1,707
Gasparilla Distance Classic	0.9	2.1	-7.4	44.0	6,014
Last Chance Bq.2 Chicagoland Marathon	-15.3	-14.4	-7.3	41.8	1,593
Honolulu Marathon	28.7	14.0	7.0	35.4	23,037
Big Island International Marathon	10.7	11.5	-7.0	30.0	2,334
REVEL Big Bear and Half Marathon	-15.1	-14.7	-6.8	45.8	7,692
Ojai 2 Ocean Marathon	-10.1	-10.1	-6.3	42.8	1,413
Five Points Of Life Marathon	-1.5	-1.7	-6.2	54.1	1,691
Deadwood-Mickelson Trail Marathon	15.9	2.9	6.1	18.9	1,789
A Run Through Time	62.3	61.3	-5.6	59.0	1,446

Appendix B Data Construction

Section 2 describes the primary data sources and estimation sample. This appendix provides additional detail on the entity-resolution procedure and the selection of the race-time variable.

Table B1 summarizes the construction of the samples used later in the paper, and the sample sizes differ across rows for specific reasons that the rest of this paragraph explains. The raw linked panel is the full web-scraped dataset after entity resolution: every finish-time observation at every marathon with runner identities assigned across races. The AKM estimation sample is smaller because the model requires each runner to appear at least twice

within a geographic region so that the runner fixed effect is identified, requires a known year of birth so that the age polynomial is defined, and trims the slowest 0.1% within each race-year to prevent outliers from dominating the residual variance. These three filters drop singletons, observations with missing year of birth, and extreme outliers, which together cut the observation count from 16.6 million to 7.7 million. The course count falls modestly because small races at which every runner is a singleton contribute no observations after the two-appearance requirement. The course speed index row is a course-level object, not an observation-level one; it reports only the number of courses retained for the published index, which are those with at least 50 AKM observations after shrinkage. The qualification analysis sample is the set of non-Boston observations at indexed courses in years with a known BQ standard. It is drawn from the full linked panel rather than the AKM sample, so singletons reenter (their BQ margin is defined without a runner fixed effect) and the observation and runner counts rise correspondingly. The cross-race additive separability pairs are a pair-level object: for each runner-race in the linked panel, I form a pair with every prior marathon by the same runner at a different course within 18 months. The runner count is lower than in the AKM sample because only runners with at least two appearances within an 18-month window can generate a pair. The Boston-destination pairs are the subset whose destination is Boston, which defines the standardized retest because the index is normalized so that $\hat{\delta}_{\text{Boston}} = 0$.

Table B1: Sample construction

Sample	Observations	Runners	Courses
Raw linked panel	16,587,223	12,157,750	1,070
AKM estimation sample	11,142,687	2,631,055	1,162
Course speed index	—	—	1057
Qualification analysis	9,625,552	5,416,443	799
Cross-race additive separability pairs	10,934,525	—	—
Boston-destination pairs	690,058	—	—

Notes: The raw linked panel is the web-scraped dataset after entity resolution. The AKM sample requires at least two appearances per regional runner, a known year of birth, and trims the slowest 0.1% within each race-year. The course speed index retains courses with at least 50 AKM observations after Empirical Bayes shrinkage. The qualification analysis uses all non-Boston observations at indexed courses with BQ standards; its course count is smaller than the index because non-US courses without a BQ-standards mapping and courses without indexed years in the qualification window drop out. Additive-separability pairs link each runner’s destination race to every prior marathon at a different course within 18 months; Boston-destination pairs restrict the destination to Boston. Samples are used as follows: the raw panel in Sections 6 and 7; the AKM sample in Sections 3 and 8; the qualification sample in Sections 5–8; the pair samples in Sections 4 and 8; and the geocoded US subpopulation in Section 5 (selection and choice analyses).

Different exercises use deliberately different “near-BQ” bandwidths, matched to each design:

Table B2: Near-threshold bandwidth conventions, by exercise

Exercise	Bandwidth	Rationale
Bunching density	± 5 min of posted standard	Spike width at the kink
Era sorting comparison	± 5 min of neutral margin	Symmetric around the target
Dissimilarity / excess sorting	$\pm 5\%$ of standard (neutral)	Proportional across demographics
Inframarginal control group	$> 15\%$ slower (neutral)	Course choice cannot affect Q_i
Choice-model pressure index	Tent peaked -8 min, width 10	Matches prior-best response peak

Because web-scraped results sometimes contain misaligned timing columns (chip times versus gun times), I also select the time variable for each race-year that minimizes within-runner prediction error across races.

B.1 Annual Summary Statistics

Table B3 reports annual panel statistics for every year in the sample. Columns show the total finisher count, the number of distinct courses, the share of rows with a known age or year of birth, the share female (conditional on a labeled gender), the count and population share of courses at least 2% faster than Boston, the count of apparent qualifiers, the share of finishers clearing their own age-gender BQ standard, and the posted and effective M18–34 BQ thresholds that governed Boston admission in the following year. The qualifying-share column reflects both the tightening BQ standard (M18-34 moves 3:10 $\rightarrow$ 3:05 $\rightarrow$ 3:00 $\rightarrow$ 2:55) and the slow drift in marathon-field composition.

B.2 Entity Resolution

I link runner records using a procedure that begins with a high-recall merge and then prunes implausible groupings using rarity and within-group contradictions. The first stage builds name-rarity tables (counts of observations sharing a given canonical first name, last name, gender, and year-of-birth bucket) so that downstream decisions can condition on how dis-

tinctive a given identifier combination is, and constructs a graph of marathons that share unusually many high-confidence linked runners; connected components of this graph define “communities” of races that recruit from overlapping runner populations. The second stage forms initial runner clusters by merging observations that share a canonical first name, last name, and gender, with year-of-birth resolved at the runner level: missing years of birth are imputed from age bands or from peer observations of the same canonical identity, and the merge requires that all observations sharing an identity have year-of-birth ranges that intersect. Country and state are imputed where partially observed, and runner-level country is propagated within the cluster.

The third stage splits clusters that look implausible. Hard-contradiction splitters separate observations within a cluster that have non-intersecting year-of-birth ranges, two finishes on the same date at different races, more than one finish at the same race in the same year, conflicting observed gender, or finishes labeled to incompatible countries. A rarity-based splitter then breaks up clusters whose name combinations are common enough that the high-recall merge is likely to have collapsed several distinct people, with rare combinations deferred to the next sub-stage. A connectedness-based splitter uses the marathon-community graph from the first stage: observations of a candidate identity that fall in disconnected race communities are split apart, since a single physical runner is unlikely to compete in entirely separate regional or competitive ecosystems. A final hard-validation sweep re-applies the contradiction tests on the post-split clusters. The output is a runner-id assignment for every observation in the panel. As a robustness check, I re-estimate the main course effects on the subsample of clusters that pass the strictest filters (rare name combinations, narrow year-of-birth windows, no high-volume aggregator flags). The resulting estimates have a Spearman correlation above 0.97 with the baseline estimates.

B.3 Time Variable Selection

Web-scraped race results sometimes contain misaligned columns, so a variable labeled “time” may correspond to gun time, chip time, or an unrelated field. For each race-year, I therefore select the time variable that minimizes within-runner mean squared prediction error, using the runner’s times in other races as the prediction benchmark. Any residual mismatch between chip and gun time is small relative to the interquartile range of estimated course effects (11 minutes).

Appendix C Robustness

I estimate six alternative specifications. The first three alter the sample restrictions: excluding trail races, restricting to high-confidence entity-resolution matches, and requiring at least three appearances per runner. The fourth drops observations within ± 10 minutes of the BQ standard, isolating identification from runners far from the strategic-effort margin. The fifth replaces each finish time with its debunched value (subtracting the mass piled up at salient minute marks within each race-year). The sixth restricts to runners who placed top 5 in their age-gender group at any race—“race-to-win” finishers for whom BQ pursuit is not the operative incentive. All six specifications yield Spearman rank correlations above 0.97 with the baseline estimates; four of the six exceed 0.99. The largest deviation arises in the top-5 specification (Spearman 0.97, mean $|\Delta FE| = 1.8$ minutes), consistent with the elite tail of finishers being a different selection of runners than the BQ-pursuing population that dominates the baseline sample.

Robustness to Strategic-Effort Contamination. A concern for the behavioral estimates of Section 5—and for any use of the runner fixed effects as neutral-course baselines (Appendix E)—is that the runner fixed effects may themselves be contaminated by strategic effort near the BQ threshold. If runners systematically exert more effort when close to qual-

ifying, their estimated baseline performance could be biased upward, which would in turn bias any residual built on it. I address this concern in two ways.

First, I re-estimate the AKM model after excluding runners within 5 or 10 minutes of the BQ cutoff. I also estimate the model on a sample restricted to inframarginal runners, defined as those whose predicted baseline performance is more than 5 minutes slower than the BQ threshold on a neutral course. In this sample, proximity to the threshold cannot affect effort. In all three cases, the estimated course effects are substantively unchanged relative to the baseline, indicating that the course-speed index is not driven by strategic effort.

Second, I re-estimate the full AKM specification after dropping all observations within ± 5 minutes of the BQ cutoff. This produces “clean” runner fixed effects identified only from observations for which runners had no immediate BQ incentive to exert differential effort. These fixed effects are highly correlated with the baseline estimates, confirming that runner-level neutral-course baselines are robust to strategic-effort contamination.

Debunched Finish Times. A further concern is that strategic bunching at BQ thresholds and round numbers mechanically inflates course fixed effects at races where bunching is concentrated. I address this by estimating a smooth counterfactual finish-time density outside an excluded bunching region around each threshold, identifying excess mass, and probabilistically reallocating bunching runners across the excluded region according to the counterfactual density. I run three asymmetric excluded windows as sensitivity: narrow $[-3, +5]$ minutes, medium $[-5, +7]$ minutes, and wide $[-7, +10]$ minutes. The debunched course effects are virtually identical to the baseline (Spearman ≥ 0.9997 for all three windows), indicating that bunching does not materially affect the estimated course speed index.

Non-Marginal Runners. To test whether course effects are driven by BQ-motivated effort, I identify runners who placed in the top K of their age-gender group at any race—runners who compete to win, not to qualify for Boston—and keep all their observations across all courses. The resulting course effects are highly correlated with the baseline (Spearman

0.94–0.98) and, if anything, show BQ-focused courses as faster, not slower, than the baseline estimates. This is inconsistent with BQ-related effort inflating the course fixed effects.

Course-Level Additivity Residuals. For each origin course r with at least 50 runner-pairs (i, r, Boston) where the same runner raced the origin course and Boston within 18 months, I compute the average log-time gap and compare it to the FE-implied gap. Let $g_{ir}^{\text{obs}} = \ln T_{i, \text{Boston}} - \ln T_{ir}$ denote the observed log-time difference (Boston minus origin) for runner i 's pair and $g_{ir}^{\text{FE}} = \hat{\delta}_{\text{Boston}} - \hat{\delta}_r = -\hat{\delta}_r$ the corresponding FE-implied difference (since $\hat{\delta}_{\text{Boston}} = 0$ by normalization). If the AKM is perfectly additively separable, the expected value of g_{ir}^{obs} is g_{ir}^{FE} and the residual is mean-zero. The *course-level additivity residual* is the course-specific average of that residual, after partialling out year-to-year temperature deviations at the origin and at Boston:

$$\hat{\Delta}_r = \mathbb{E}_{i \in \mathcal{P}_r} \left[(\ln T_{i, \text{Boston}} - \ln T_{ir}) - (-\hat{\delta}_r) \right] = \mathbb{E}_{i \in \mathcal{P}_r} [\ln T_{i, \text{Boston}} - \ln T_{ir}] + \hat{\delta}_r, \quad (8)$$

where $\mathcal{P}_r$ is the set of runners who raced both course r and Boston within the 18-month window. Reported in minutes at a 3:20 reference pace, $\hat{\Delta}_r$ measures how much slower (or faster) runners actually performed at Boston than the origin course's fixed effect would have predicted. A perfectly additively separable index would have $\hat{\Delta}_r = 0$ at every course; positive values indicate the FE overstated the origin course's speed (runners were slower at Boston than the FE implied), and negative values indicate the FE understated it.

Figure C2 plots $\hat{\Delta}_r$ against the origin-course FE for 345 courses with $|\hat{\delta}_r| \leq 25$ minutes. Most courses sit inside a ± 5 minute reference band around zero, confirming that the FE index is additively separable to within a few minutes course by course. Most of the small positive shift in the mean (+2.5 minutes) reflects a Boston-specific slowdown common to all origin courses—travel fatigue, the April race date, and the Boston course's own well-known difficulty—that the FE normalization to Boston cannot absorb at the individual-runner level. The flat BQ.2 courses show near-zero gaps, confirming that their fast fixed effects accurately

predict Boston performance.

Table C1 summarizes Spearman rank correlations across all specifications. Figure C1 plots all specifications for key races.

Table C1: Spearman rank correlations with baseline course fixed effects across robustness specifications. Mean $|\Delta\text{FE}|$ reports the average absolute change in estimated course speed (minutes at 3:20 pace) relative to the baseline.

Specification	N Races	Spearman ρ	Mean $ \Delta\text{FE} $ (min)
<i>Sample restrictions</i>			
Exclude trail races	731	0.999	0.2
High-confidence ER only	721	0.953	2.3
Min 3 appearances	799	0.998	0.4
<i>Strategic-effort robustness</i>			
Excl. BQ margin ± 10 min	804	0.994	0.8
<i>Debunched finish times</i>			
Debunched (medium $[-5, +7]$)	815	1.000	0.1
<i>Non-marginal runners</i>			
Top 5 by age group	804	0.969	2.1

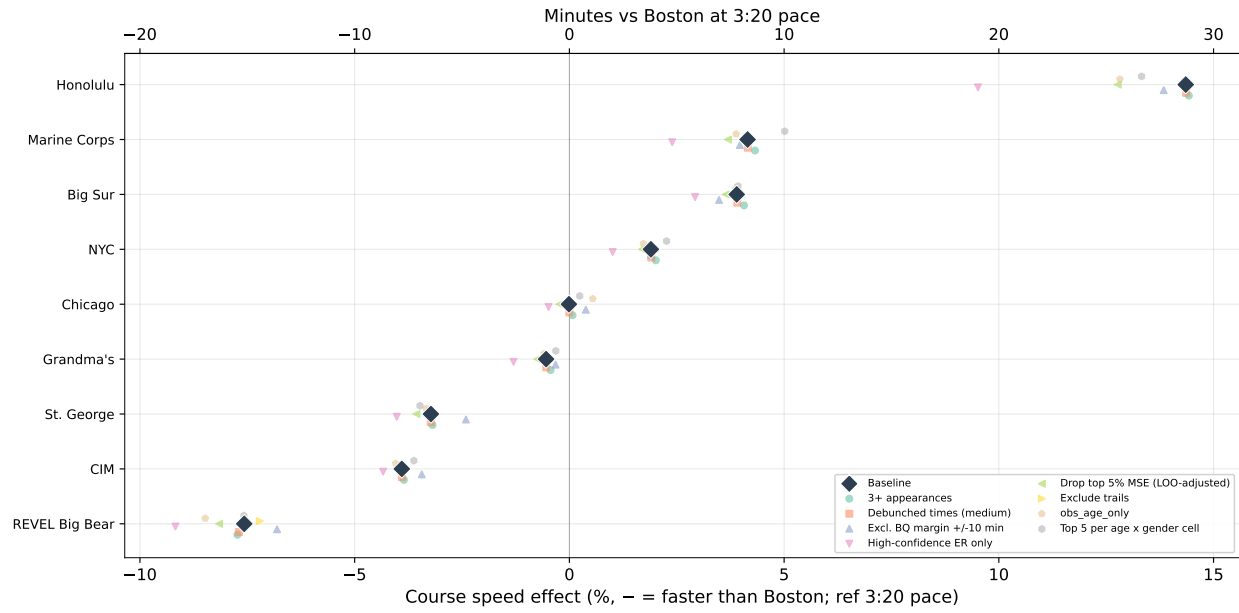


Figure C1: Course speed effects for key races across robustness specifications. Diamond: baseline AKM estimate. Small markers: alternative specifications including sample restrictions, BQ-margin exclusions, debunched finish times, and non-marginal runners.

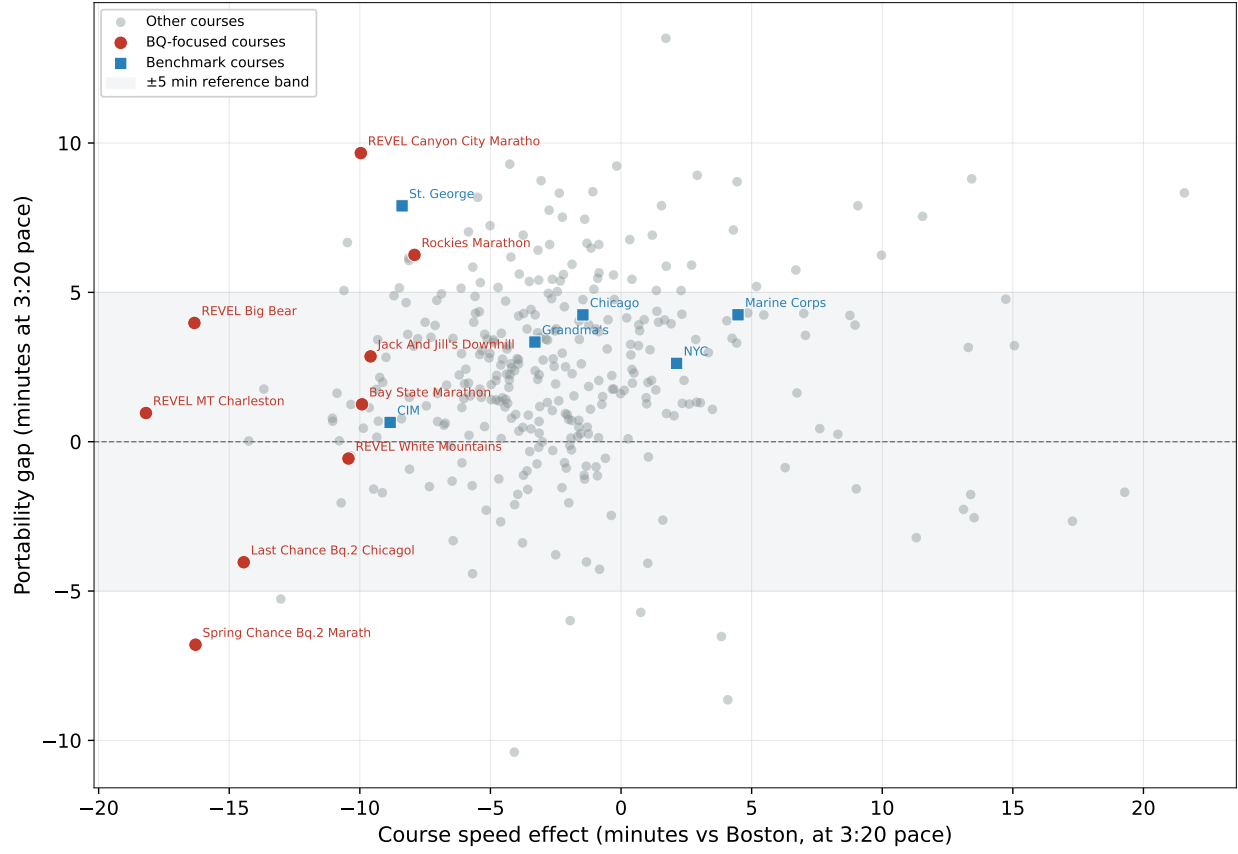


Figure C2: Course-level additivity residual $\hat{\Delta}_r$ (equation (8)) plotted against the origin-course FE. Each point is a qualifying course with at least 50 same-runner pairs who subsequently raced Boston within 18 months (weather-adjusted, restricted to $|\hat{\delta}_r| \leq 25$ minutes). Perfect additive separability implies $\hat{\Delta}_r = 0$; positive values mean the FE overstated the origin course’s speed (runners slowed more at Boston than expected), negative the reverse. Most courses sit inside the ± 5 minute reference band. Labeled outliers: Brooklyn Marathon (-18 min) has known course-measurement history (alternative paths through Prospect Park in different years), so Brooklyn times look slower than they were and Boston appears artificially fast by comparison. Mt Hood ($+11$ min) and Clearwater ($+10$ min) are small, goal-race courses whose co-racing subsample is heavily peaked—runners hit peak performance at the goal race and regress at Boston. LA Marathon ($+8$ min) has the highest precision in the figure ($N = 6,128$) and its FE appears to underweight the Boston-specific heat slowdown because LA itself is a mild-weather race. Spring Chance BQ.2 sits just below zero, consistent with its BQ-focused design producing a small additional peaking effect at the origin that does not fully propagate to the Boston run.

Robustness of the Misallocation Rate Series. The headline misallocation series in Figure 9 counts race-results (so a runner who qualifies on three different fast courses in the same year contributes three observations) and restricts to AKM-covered courses (those with

at least 50 observations after shrinkage, the only courses for which $\hat{\delta}_r^{shr}$ is defined). Figure C3 reports two alternatives. The first counts unique runners rather than race-results: a runner with multiple qualifying races in year t contributes one to the denominator and one to the numerator if any of those races was course-dependent. The second expands the denominator to include qualifiers at non-AKM-covered courses, treating those qualifiers as not course-dependent (a lower-bound on the true rate, since some non-AKM courses are likely fast). Both alternatives track the headline series closely: the unique-runner adjustment lowers the level by 1–2 percentage points without changing the trend, and the broader-sample lower bound runs about 3–4 percentage points below the headline. The qualitative pattern—a roughly doubling of the misallocation rate between 2004 and 2019—is robust to both choices.

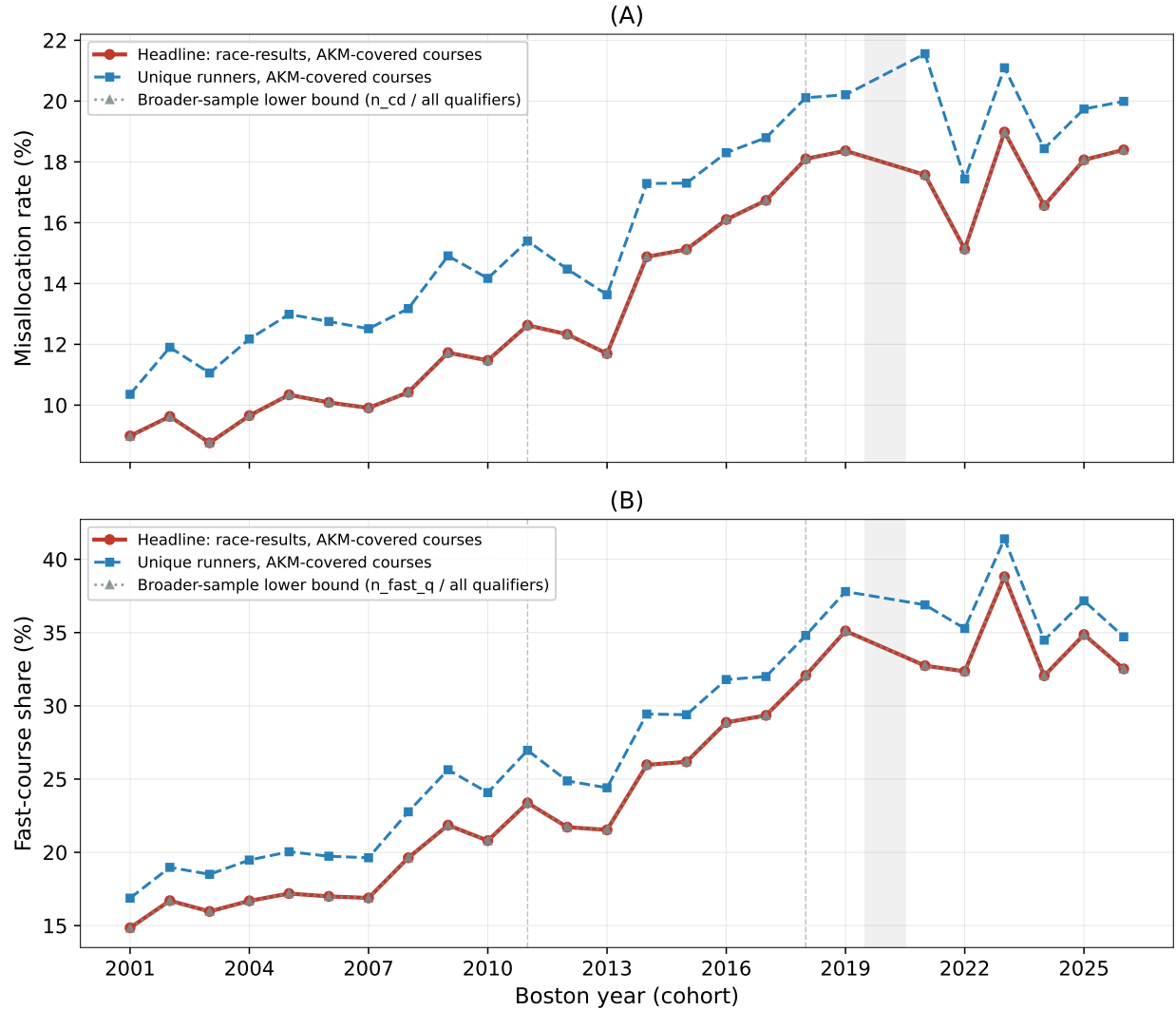


Figure C3: Robustness of the misallocation-rate and fast-course-share series, Boston 2001–2026 cohorts. The headline series (red, race-results among AKM-covered finishers) is the same as in Figure 9. The unique-runner series (blue) collapses each runner to a single observation per Boston cohort, with course-dependent and fast-course indicators set to one if any of that runner’s qualifying-window races meet the criterion. The broader-sample lower bound (gray) expands the denominator to all qualifiers (including those at non-AKM courses) but cannot classify non-AKM-course qualifiers as course-dependent, so it is a lower bound on the true rate. Boston 2000 excluded (partial cohort); cancelled Boston 2020 shaded; vertical dashed lines mark the first cohort under each new standard (Boston 2013 for 3:05; Boston 2020 for 3:00; Boston 2026 for 2:55 under-60).

Exhibits Supporting the Main-Text Series. The following exhibits supplement Sections 7, 5, 6, and 8: the qualifying-menu mechanics, the near-BQ versus inframarginal sorting series, the estimand-reconciliation and predictable-pressure tables, the excess-mass detail,

the full decomposition chains by cohort with the alternative-baselines grid, the course-level over-attraction ranking, and the displacement comparison across every index and rule.

Table C2: Reconciling the sorting magnitudes: each estimate, its estimand, its population, and the channels it includes.

Estimate	Value	Estimand	Population	Channels
Excess fast-course share, near-BQ	+9.8 pp	Stock: fast-course share of near-BQ finishers minus the era’s full field	Near-BQ finishers (neutral ± 5 min), 2013–2019	All BQ-related sorting (venue choice + correlated effort)
Announcement difference-in-differences	+4.2 pp (SE 0.3)	Change: differential fast-course choice of near-BQ runners after the 2011 tightening, net of year and runner controls	US runners with geocoded home	Venue choice (demand) response to the announcement
Choice-model counterfactual	+3.3 per 1,000	Flow in qualification units: marginal qualifications produced by the pressure-induced choice shift, course menu fixed (partial equilibrium)	Near-BQ runner-occasions in the choice sample	Pressure $\times$ speed channel of the conditional logit only
Misallocation rate (screen side)	16% $\rightarrow$ 24%	Stock: share of apparent qualifiers whose qualification depends on the course they ran (2004 $\rightarrow$ 2019 cohorts)	Apparent qualifiers at indexed courses	Outcome of all channels combined, screen side

Table C3: Selection on predictable pre-race state: pressure, venue speed, and the announcement difference-in-differences.

	(1)	(2)	(3)	(4)	(5)
Dep. variable:	Course speed (log FE)	BQ pressure (index 0–1)	Fast course (pp)	Qualifies (pp)	Boston take-up (pp)
BQ pressure	-0.0151		11.19	24.27	29.37
Chosen-course speed (log FE)		-0.5801			
Near BQ (± 5 min)	0.0059		-5.47	-9.22	-6.35
Near BQ $\times$ post-2011	-0.0024		3.18	0.20	-3.79
Post-2011	-0.0046	-0.0051	5.81	-1.21	-1.59
Log distance to course	-0.0002	0.0039	0.71	0.35	0.48

Table C4: Polynomial excess-mass estimates at the BQ threshold and round-number placebo, by course type and era.

	Pre-2012	2012+
<i>Excess mass at the BQ threshold</i>		
All courses	2.14	1.36
Fast courses	3.15	1.85
Medium courses	2.08	1.18
Slow courses	1.49	1.06
<i>Excess mass at the 4:00 round-number placebo</i>		
All courses	1.83	1.96
Fast courses	2.05	1.71
Medium courses	1.86	2.07
Slow courses	1.66	1.80

Table C5: Channel contributions to the 2019 misallocation rate (pp) under alternative assignment baselines and path orderings. Sorting is invariant to the ordering; reordering reallocates only between the threshold and supply channels. Residence-subsample rows are comparable to each other, not to the full-sample rows.

Baseline	Ordering	Threshold	Supply	Sorting
Random (field shares)	Threshold first	2.3	5.1	5.5
Random (field shares)	Menu first	3.7	3.7	5.5
Inframarginal ($ \text{margin} > 20$ min)	Threshold first	2.1	4.5	6.6
Inframarginal ($ \text{margin} > 20$ min)	Menu first	3.4	3.3	6.6
Inframarginal ($ \text{margin} > 15$ min)	Threshold first	2.1	4.7	6.4
Inframarginal ($ \text{margin} > 30$ min)	Threshold first	2.1	4.4	7.0
Inframarginal, gender-conditioned	Threshold first	2.1	4.6	6.6
Random, residence subsample	Threshold first	2.6	5.1	7.2
Gravity (distance decay), residence subsample	Threshold first	2.6	4.8	7.8
Gravity (distance decay), residence subsample	Menu first	4.1	3.3	7.8

Table C6: Displacement under alternative indices and rules (fixed field, US 2018–19).

Policy	Displaced (%)	From fast courses (%)	Runners out	Runners in
Common threshold (status quo)	0.0	–	0	0
Behavior-adjusted (headline)	13.9	87.5	11,876	11,876
AKM physical (naive ψ , starting point)	14.6	88.0	12,477	12,477
Clean AKM (excluded-margin)	14.3	87.7	12,184	12,184
Routine-attempt (visitor control)	12.1	83.5	10,336	10,336
Reverse-pair (symmetric)	14.9	85.2	12,722	12,722
Observable-based (terrain)	10.5	61.7	8,947	8,947
Conservative shrinkage ($k = 1.645$)	11.9	91.0	10,195	10,195

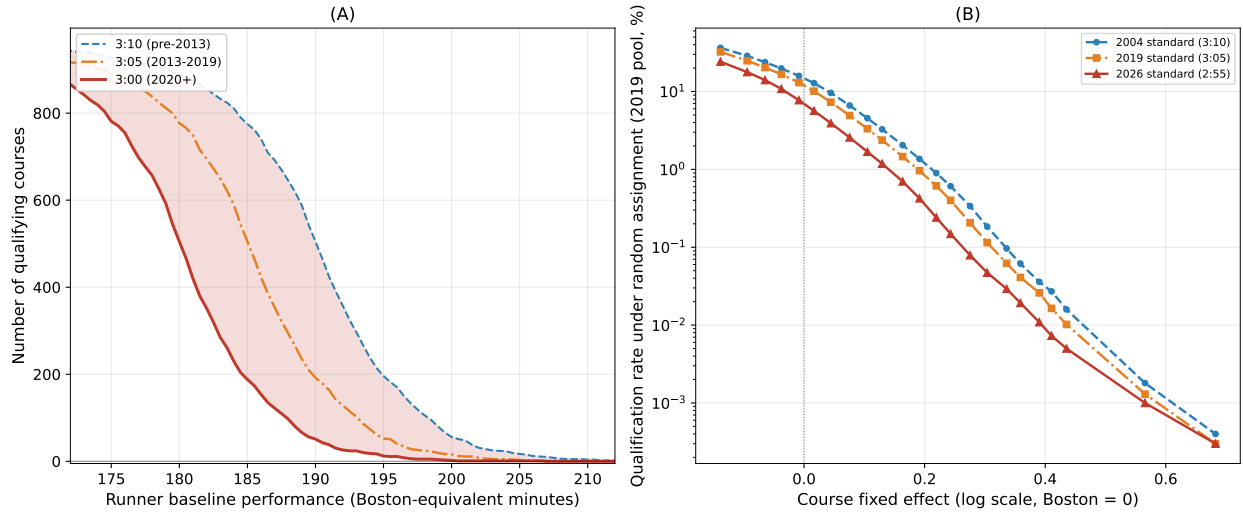


Figure C4: Qualifying mechanics. Panel A: the number of indexed marathons at which a runner of given neutral ability can qualify, under three historical standards. Panel B: qualification rates by course speed under the same three standards.

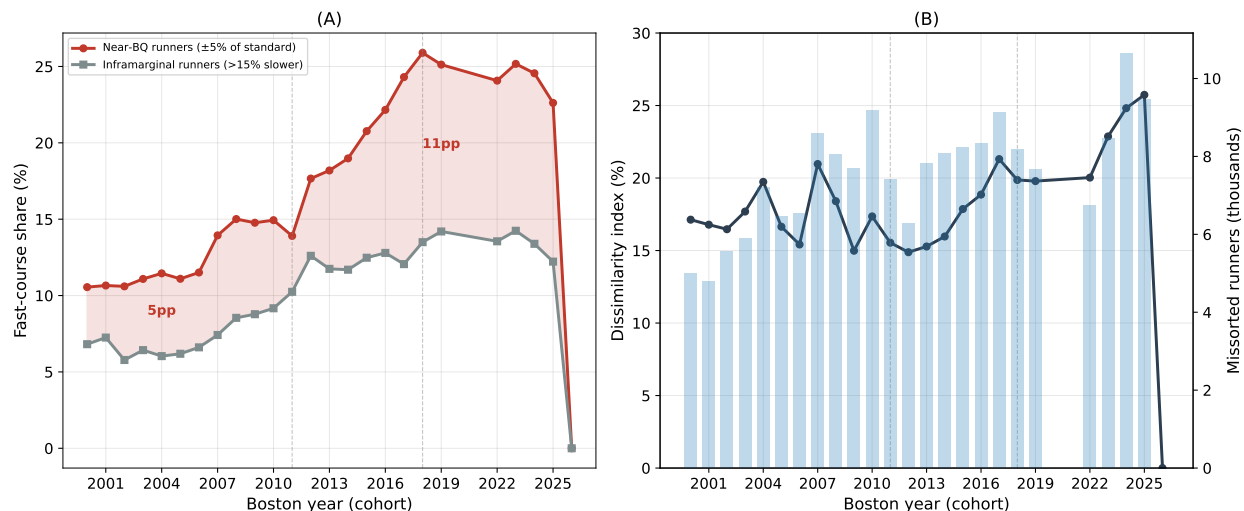


Figure C5: Near-BQ vs. inframarginal fast-course shares (Panel A) and the implied excess sorting (Panel B), 2000–2025. Under the assumption that inframarginal runners supply the no-incentive counterfactual distribution, roughly one in five near-BQ runners is at a course they would not otherwise have chosen.

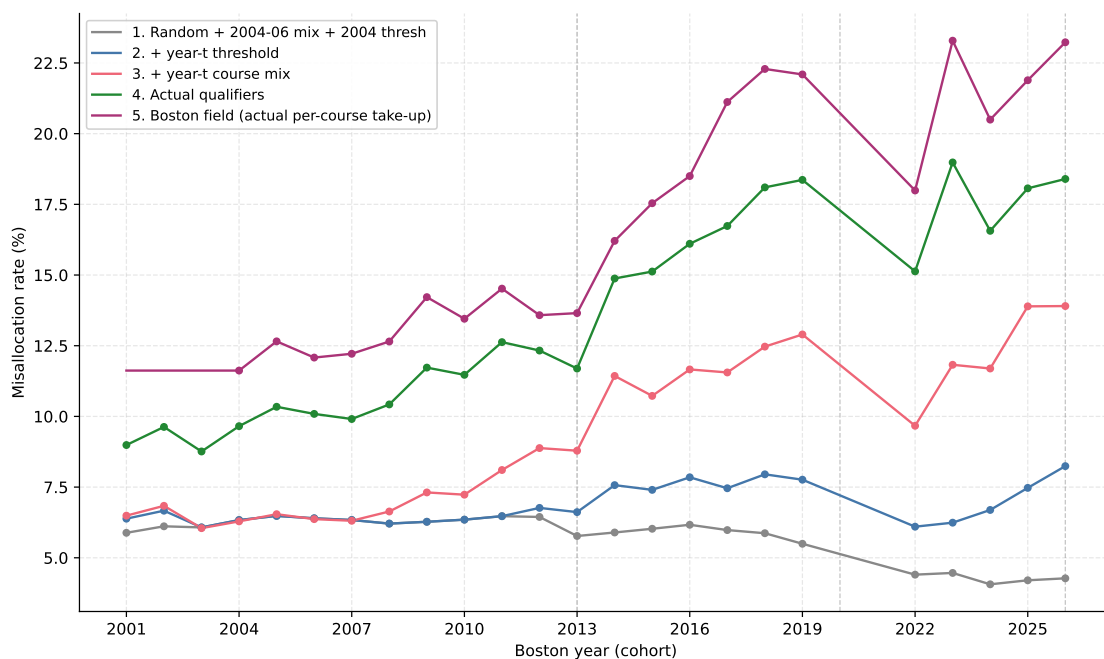


Figure C6: Counterfactual decomposition of the misallocation rate, Boston 2001–2026 cohorts. Step 1 random baseline; step 2 adds the cohort threshold; step 3 adds the cohort course menu; step 4 is actual; step 5 weights by take-up. Vertical lines mark the first cohort under each new standard.

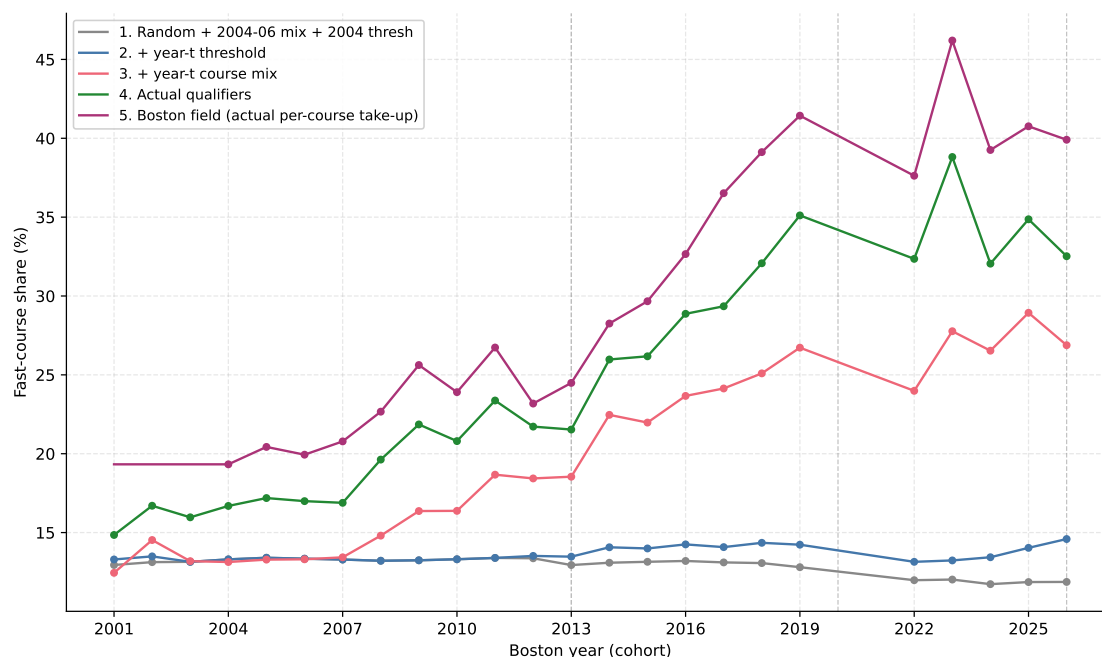


Figure C7: The same decomposition for the fast-course share among apparent qualifiers (steps 1–4) and Boston participants (step 5).

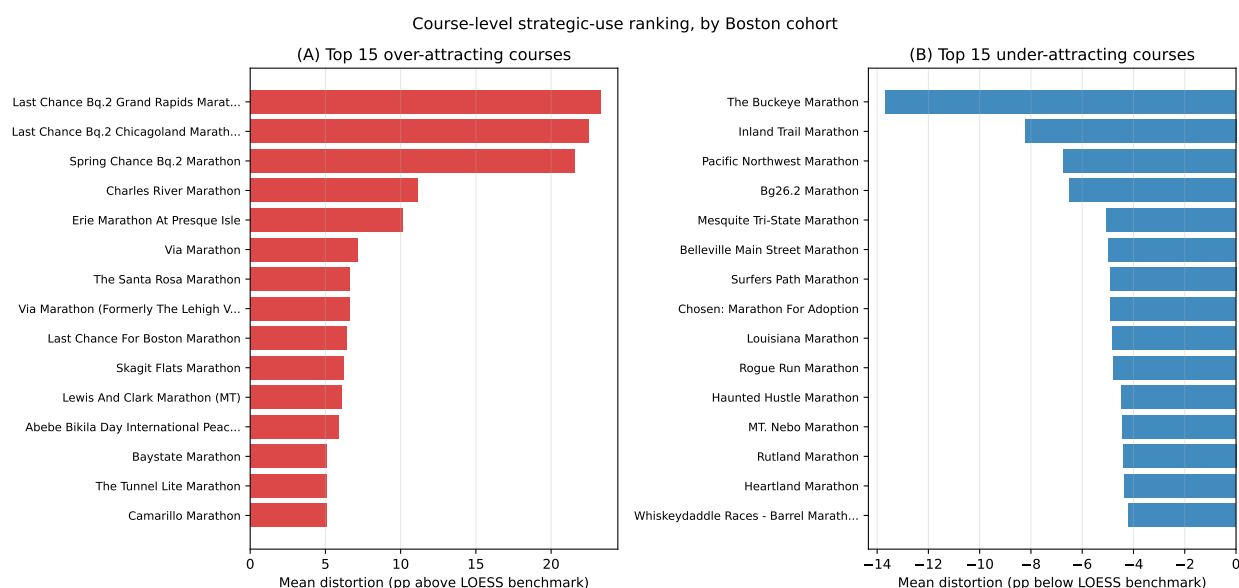


Figure C8: Top over- and under-attracting courses, ranked by cohort-weighted deviation from the smooth speed benchmark. Purpose-built BQ products lead the over-attractors.

The Full Corrected-Index Suite. Table C7 reports the complete seven-index suite behind the four-index main-text version (Table 2): the three variants demoted from the

main text are the clean re-estimate (excluding near-threshold observations), the reverse-pair symmetric estimator, and the conservative shrinkage rule. The displacement band quoted throughout (10.5–14.9%) spans all seven.

Table C7: The full course-speed index suite: construction, what each neutralizes, and the implied fixed-field displacement (US 2018–19 qualifiers, field size 85,466).

Index	Neutralizes	Displaced (%)
Behavior-adjusted (headline), eq. (4)	Course-level peaking of strategic users	13.9
Naive AKM ψ (starting point)	—	14.6
Clean AKM (excluded margin)	Strategic effort near the threshold	14.3
Routine-attempt	Visitor/goal-race composition	12.1
Reverse-pair (symmetric)	Direction-specific order effects	14.9
Observable-based (terrain)	All non-physical variation (and some physical)	10.5
Conservative shrinkage ($k=1.645$)	Estimation noise (zeroes imprecise courses)	11.9

Appendix D AKM Diagnostics

Following Card, Heining and Kline (2013), I assess the AKM decomposition on two dimensions: year-to-year stability of course effects within each race, and nonseparability of runner and course effects. Additive separability (Section A.4) already provides the mover-style cross-venue validation that is typically the third CHK diagnostic. These exercises are intended as stress tests of the additive approximation rather than as alternative estimators of the main course index.

Within-Course Year-to-Year Stability. If the estimated course effect captures a persistent feature of the race, the year-to-year variation in finish times at a given course, net of common year trends and runner composition, should be small relative to the cross-course

variation in the index. I measure this stability using the standard deviation of race-year mean residuals within each course. After the AKM, for each race-year cell I compute $\bar{e}_{r,t} = \text{mean}(\hat{\varepsilon}_{i,r,t})$; the within-course across-year SD of $\bar{e}_{r,t}$ captures year-to-year course variation net of the common year FE.

Across the 691 courses with at least three years of data, the median within-course across-year SD is 2.8 minutes (at a 3:20 pace), the 95th percentile is 6.7 minutes, and the mean is 3.3 minutes. These are small compared with the cross-course SD of the permanent course effect itself (on the order of 20–30 minutes), implying that $\hat{\delta}_r$ is well identified as a time-stable course attribute. Figure D1 plots this within-course across-year SD against the permanent course effect for each race, highlighting benchmark courses. Marine Corps (1.9 min), St. George (2.2 min), and Honolulu (2.2 min) are among the most stable despite spanning a wide range of course speeds; Boston itself is at 3.7 minutes, reflecting genuine year-to-year variation in weather and field composition.

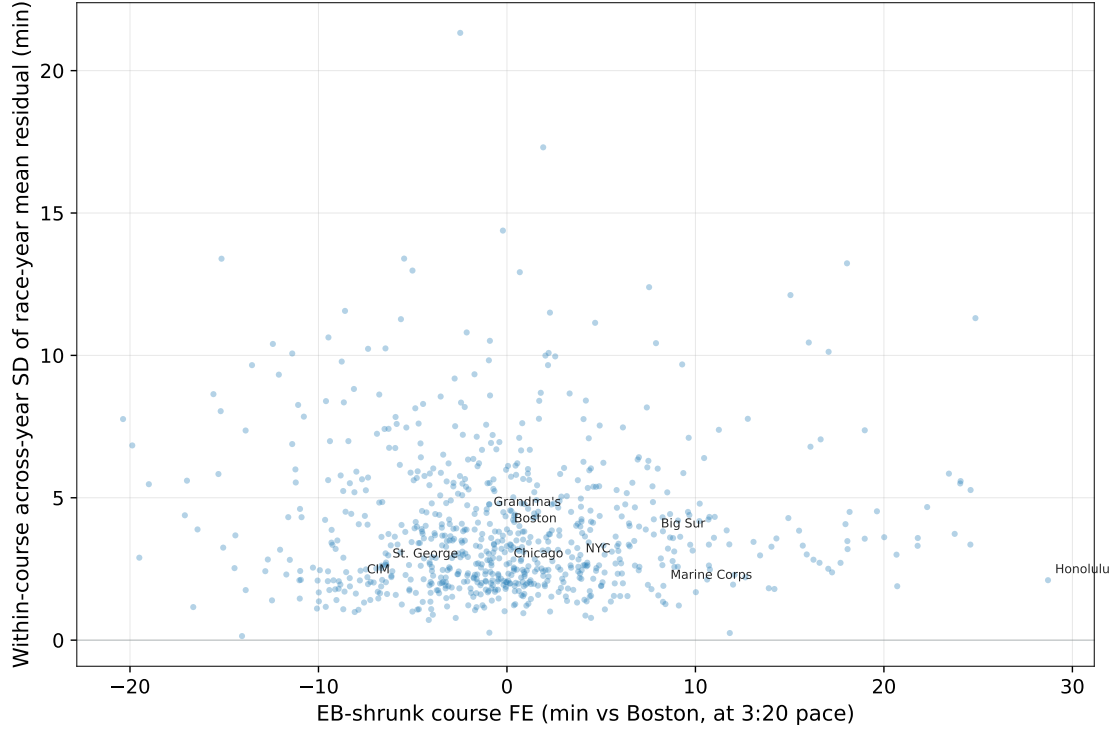


Figure D1: Within-course year-to-year stability of the AKM course effect. Horizontal axis: EB-shrunk course FE $\hat{\delta}_r^{shr} = w_r \hat{\delta}_r$ in minutes vs Boston at 3:20 pace ($|\hat{\delta}_r^{shr}| \leq 25$ min plus benchmarks always retained). Vertical axis: standard deviation of race-year mean residuals within that course, net of the common year FE, the runner and course FEs, the age-gender polynomial, and year-to-year deviations in morning temperature. Humidity, wind, and precipitation are not in the AKM and therefore contribute to this SD. The median across-year SD is 2.8 minutes and the 95th percentile is 6.7 minutes, small relative to the cross-course FE range. Empirical Bayes shrinkage has already pulled the volatile courses toward zero: the weights range from 0.73 at the noisiest course (Waterloo, which has only 3 years and 59 observations) to 0.999 at Boston, Chicago, and NYC; the largest shrinkages are between 10 and 27 percent. The remaining labeled high-SD outliers have thin samples (Waterloo, Groundhog, Loonies Midnight, and Bay of Fundy International each have fewer than ten race-years and under 400 total finishers, so the race-year mean residual is noisy even after shrinkage) or weather-sensitive designs (REVEL Rockies is a high-altitude downhill race with documented year-to-year variation in wind and precipitation beyond what temperature alone captures). None of these represent identification failures of the AKM; they are cases where either the cell sample is thin or the non-temperature weather draw is genuinely volatile.

Nonseparability. Under the additive AKM structure, the course effect shifts all runners' log times by the same amount, so mean residuals should be approximately zero in every runner-ability $\times$ course-speed cell. To test this, I assign each observation to a runner-FE decile and a course-FE decile and compute the mean residual in each of the 100 cells. I then

two-way demean to isolate the runner $\times$ course interaction: $\hat{\eta}_{rc} = \bar{e}_{rc} - \bar{e}_r - \bar{e}_c + \bar{e}...$. Under the additive model, $\hat{\eta}_{rc} \approx 0$ for all cells.

Figure D2 displays the interaction heatmap. The largest cell deviation is 0.61 log points $\times 100$ (approximately 0.6% of finish time), and the mean absolute deviation is 0.10%. The interaction terms do exhibit some systematic structure, but it is concentrated in extreme runner-course cells and modest relative to the main course effects. In particular, the largest deviations occur in the tails of the runner and course distributions rather than in the middle of the support where the BQ-margin analysis is concentrated. These results indicate that the additive structure provides a useful approximation for road marathon performance, while leaving room for modest tail heterogeneity.

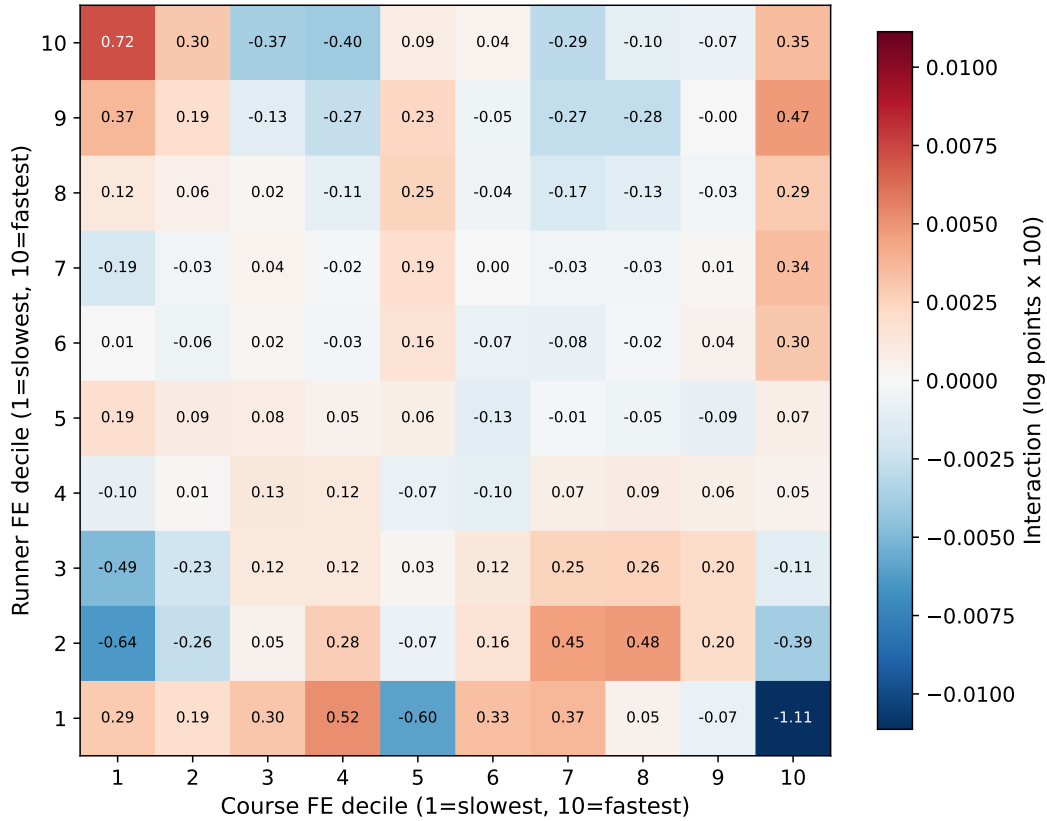


Figure D2: Nonseparability test: two-way demeaned interaction residuals by runner-FE decile (rows) and course-FE decile (columns). Values are log points $\times 100$ (approximately percent). Under the additive AKM model, all cells should be near zero.

Appendix E Additional Bunching Diagnostics

This appendix collects supporting exercises for the demand-side evidence in Section 5: two bunching checks, a cross-margin pressure-index calibration, and the age-group notch analysis. The main-text figure 5 establishes that bunching is BQ-specific, that the spike tracks the qualifying standard as it moves, and that excess bunching on fast courses has grown over time. Here I report two further checks: whether the bunching could reflect the underlying ability distribution rather than behavior, and a granular version of the generic-versus-BQ comparison.

The AKM runner fixed effects, converted into predicted finish times on a neutral course, imply a baseline performance distribution that is smooth through salient round-number thresholds (Figure E1, Panel A). Actual finish times, by contrast, display pronounced spikes at half-hour and hour marks (Panel B). This contrast indicates that the bunching documented in Section 5 reflects behavior rather than irregularities in the underlying ability distribution.

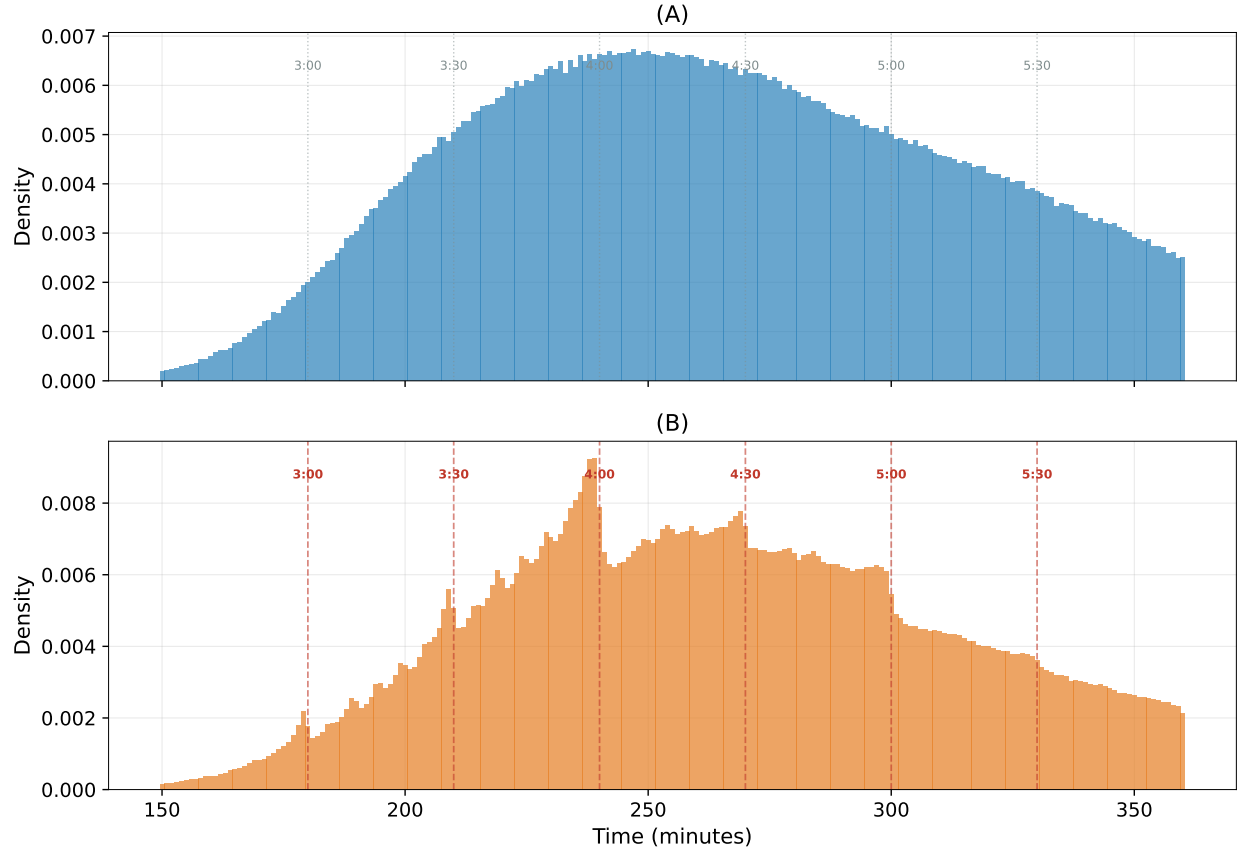


Figure E1: Baseline runner performance and actual finish times. Panel A plots the distribution of predicted neutral-course finish times. Panel B plots the distribution of actual finish times. The baseline distribution is smooth, whereas actual finish times bunch at salient thresholds.

Figure E2 provides additional detail on the contrast between generic round-number bunching and BQ-specific sorting. The top row examines fast-course share and finish-time density around the 4:00 mark (excluding runners for whom 240 minutes is a BQ threshold). The fast-course share is essentially flat across 4:00 (25.0% just above the mark, 26.4% just below), whereas it rises sharply at the BQ threshold (34.7% just above, 38.6% just below). The moving-BQ exercise in the main text (Figure 5) summarizes the same pattern.

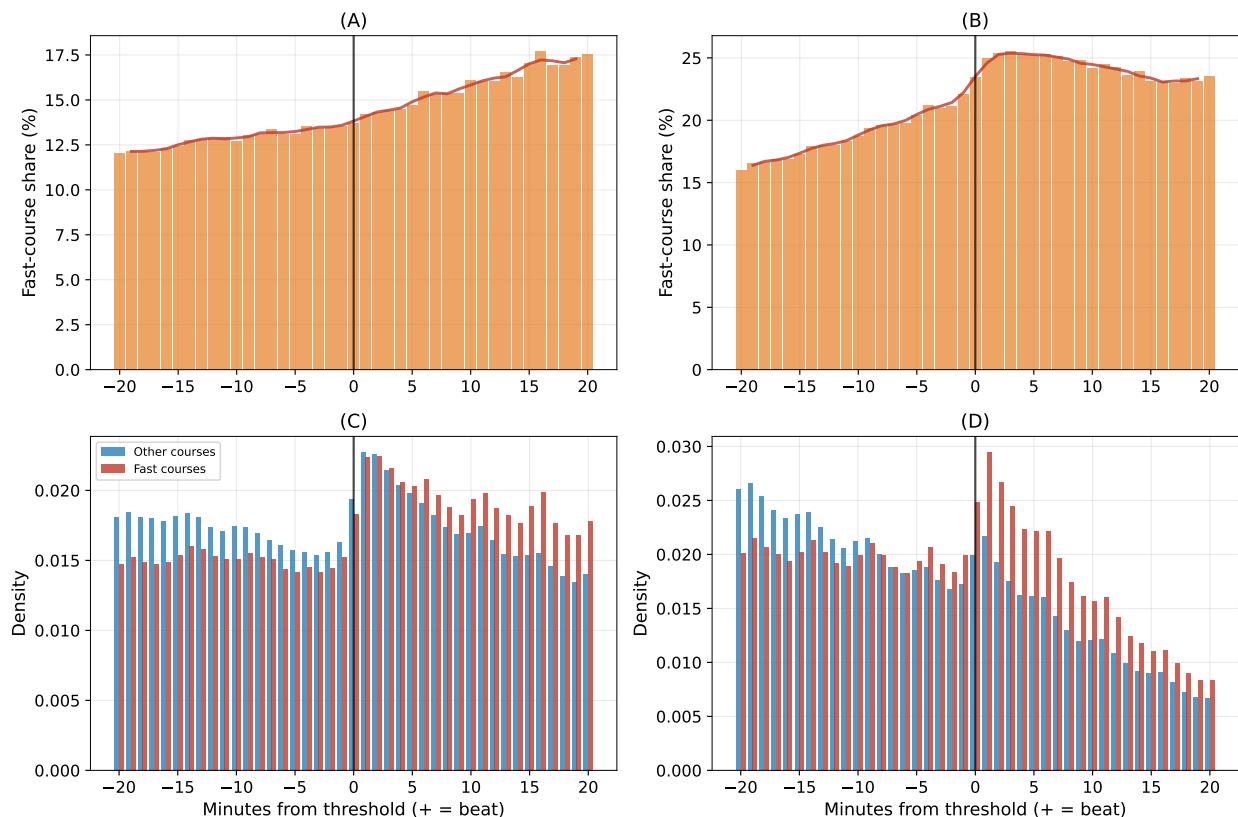


Figure E2: Generic round-number bunching and BQ-specific sorting. The top row examines the 4:00 mark, excluding runners for whom 240 minutes is the BQ threshold. The bottom row examines the BQ threshold. Sorting into fast courses is limited at the generic round number but pronounced at the BQ cutoff.

A Common Pressure Index Across Behavioral Margins. A constructed pre-race BQ-pressure index orders the sorting and bunching moments in the predicted direction in a cross-channel leave-one-out calibration, while the peaking-by-origin cross-section is dominated by course composition (fast, competitive origins draw consistent runners who fade less), even though the individual qualifier-peaking effect is solidly identified in Table A6. Because one of the three moments does not aggregate cleanly, the main text cites this calibrated consistency check only as corroboration; the underlying estimates are in the replication archive.⁶

⁶Full simulated-likelihood estimation of the joint model, with standard errors on the single-index restriction, is a natural next step and is beyond this paper.

E.1 The Age-Group Notch: Two Margins, Both Bounded

Qualifying standards relax discontinuously at five-year age boundaries, and the applicable category is determined by age *on Boston race day*—predetermined relative to any qualifying attempt—so threshold pressure is relieved exactly at aging-up. Two margins could respond, and the data bound both. The venue-choice margin cannot be estimated here: the data record year of birth but not birth date, and fall qualifiers map to a Boston nearly two years later, so a runner’s Boston-day age is identified only to within about two years, and the boundary-adjacent cells—precisely where the response would appear—are the cells that cannot be classified with certainty. The attempt-timing margin can be tested despite the fuzziness: a runner who ages up between Boston year B and $B+1$ can delay a qualifying attempt past the mid-September window boundary so that it counts under the easier standard, and that retiming is detectable at the year level. The test returns a precise null: among near-margin runners who surely age up, the shift in post-boundary attempt share is 0.8 percentage points, with a confidence band of roughly ± 1 points after accounting for classification dilution, and a far-from-margin placebo attributes the remaining small contrasts to age-trend artifacts rather than incentives. The notch design therefore neither supports nor refutes the venue-choice prediction, the retiming channel is empirically absent, and no exhibit in the paper rests on either margin.

Table B3: Annual Summary Statistics – US Marathon Panel

Year	Finishers	Courses	Age%	F%	Fast courses	Fast%	App. Q.	BQ%	BQ _{M18} (min)	Eff. _{M18} (min)
2000	295,449	210	87	37	30	7.1	29,043	9.8	190	190.0
2001	272,053	217	92	36	28	7.7	24,597	9.0	190	190.0
2002	314,800	183	94	39	24	6.5	33,039	10.5	190	190.0
2003	326,159	205	93	39	28	7.0	31,423	9.6	190	190.0
2004	357,300	243	99	40	35	7.1	29,507	8.3	190	190.0
2005	380,656	262	98	40	36	7.0	33,924	8.9	190	190.0
2006	396,062	285	99	40	36	7.3	41,338	10.4	190	190.0
2007	401,342	296	99	39	41	8.5	41,355	10.3	190	190.0
2008	414,144	312	95	41	52	9.2	44,320	10.7	190	190.0
2009	460,915	313	95	40	50	9.5	53,471	11.6	190	190.0
2010	493,324	304	100	41	51	10.0	52,786	10.7	190	190.0
2011	491,368	337	98	41	59	10.4	54,228	11.0	190	188.8
2012	473,579	367	90	42	67	12.2	35,550	7.5	185	185.0
2013	529,993	379	94	43	70	12.1	47,967	9.1	185	183.4
2014	544,236	386	98	44	73	12.1	49,016	9.0	185	184.0
2015	505,689	366	97	44	77	13.1	42,583	8.4	185	182.5
2016	496,042	366	94	44	78	13.3	38,378	7.7	185	182.8
2017	469,389	341	99	44	77	13.5	37,430	8.0	185	181.6
2018	448,391	346	93	44	82	14.6	42,922	9.6	185	180.1
2019	438,014	330	93	42	78	14.7	35,747	8.2	180	178.3
2020	40,652	41	98	43	13	11.6	2,138	5.3	180	172.2
2021	208,172	218	96	41	52	17.4	23,788	11.4	180	180.0
2022	354,366	277	96	41	62	14.5	40,571	11.4	180	180.0
2023	406,683	300	95	40	69	15.0	43,228	10.6	180	174.5
2024	478,421	405	99	40	78	14.5	45,219	9.5	180	173.2
2025	468,708	329	90	40	65	12.8	47,398	10.1	175	170.4

Panel covers race years 2000–2025; 2020 reflects near-total race cancellation due to COVID-19. Fast courses defined as $\hat{\delta}_r < \log(0.98) \approx -0.0202$ (approximately 2% faster than Boston). Apparent qualifiers are finishers whose actual time cleared their age-gender cell’s nominal BQ standard in force for the following April’s Boston Marathon. BQ_{M18} is the nominal M18-34 standard; Eff._{M18} is the effective (field-size-adjusted) M18-34 cutoff for the following April’s Boston field. The most recent race-year row therefore pairs with the next Boston cohort; the 2025 row pairs with Boston 2026, whose admission cutoff was announced on September 23, 2025 as 2:50:26 (BQ 2:55:00 minus a 4:34 admission buffer). The BAA reported 33,249 qualifier applications, 24,362 accepted, and 8,887 rejected for the 130th Boston (Boston Athletic Association, 2025*b*). 2003 coverage reflects a scraping gap being filled in ongoing data-collection work.

Appendix F The Course-Choice Model

A conditional-logit model estimates venue selection directly. Each choice occasion is one non-Boston US marathon finish by a US-residence runner with at least one prior observed race; the choice set is the course actually run plus nine other US courses drawn uniformly at random from those active in the same calendar year (under uniform sampling of alternatives the conditional logit remains consistent), estimated on a random subsample of 300,000 occasions contributed by roughly 238,000 distinct runners. Utility falls in a fixed long-distance travel cost and rises in the interaction of pre-race BQ pressure with course speed; the key coefficient is $\hat{\rho} = -6.10$ on pressure $\times$ speed, robust to entering distance as a long-trip dummy, continuously, or not at all (Table F1; standard errors clustered by runner). The policy-relevant translation is not the average shift in chosen-course speed, which is mechanically small, but the change in *qualification probability* the pressure-induced choice produces: switching off the pressure channel and re-solving the model, the response is concentrated among runners within a few minutes of the threshold and vanishes in the tails, adding about -11.7 marginal qualifications per 1,000 near-BQ runners in partial equilibrium with the course menu held fixed. One measurement note: the pressure variable is built from prior-best margins neutralized with $\hat{\delta}$, so estimation error enters both sides; results are robust to using raw prior times relative to the standard.

Table F1: Course-choice conditional logit: venue-selection response and travel cost.

	Coefficient	Clustered SE	z
Travel > 100 miles	−3.4841***	0.0068	−513.2
Course speed (log course FE)	3.8501***	0.0559	68.9
Pre-race qualifying probability	0.6920***	0.0311	22.2
BQ pressure $\times$ course speed	−6.1005***	0.1776	−34.4
BQ pressure $\times$ qualifying probability	−1.0603***	0.0510	−20.8

Notes: McFadden conditional logit estimated by maximum likelihood. Each choice occasion is one non-Boston US marathon finish by a US-residence runner with at least one prior observed race; the choice set is the course actually run plus up to 9 other US courses drawn uniformly at random from the courses active in the same calendar year (sampled-alternatives estimator, consistent under uniform sampling). Standard errors are cluster-robust at the *runner* level (score-based sandwich with CR1 correction; 221,682 runner clusters, 299,938 occasions); runners contribute many occasions, so observations are not independent across occasions of the same runner. BQ pressure is occasion-constant, so its main effect differences out and it enters only through the interactions. Log-likelihood −492,046. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix G Course Speed Index

Table G1 reports the estimated course speed effect for all 1057 indexed marathons, sorted from fastest to slowest. Course effects are expressed as the log difference relative to Boston (the Empirical Bayes–shrunk AKM estimate), along with the percentage equivalent and minute conversions at two reference paces: 3:00 (the current M18–34 qualifying standard) and 3:20 (approximately the median Boston qualifier). Negative values indicate courses faster than Boston. The full table is available as a CSV file in the replication archive.

Table G1: Course Speed Index (Top 25 and Bottom 10). The column “Obs. Boston slow.” reports the raw mean $\ln T_{\text{Boston}} - \ln T_{\text{origin}}$ in minutes at 3:20 pace from same-runner Boston-destination pairs (at least 50 pairs required; “—” otherwise). Under perfect additive separability, this number should equal the negative of the Min (3:20) column: a course 18 min faster than Boston should produce runners who are +18 min slower at Boston. Small gaps are consistent with measurement noise or Boston-specific factors; large gaps can indicate peaking (observed slowdown exceeds the FE prediction when the origin is a goal race) or course-measurement issues at the origin.

Marathon	Location	% vs Boston	Min (3:20)	Min (3:00)	Obs. Boston slow. (min)	Race-Years
<i>25 Fastest Courses</i>						
Inland Trail Marathon	Ohio	-10.2	-20.4	-18.3	+22.8	5
The Buckeye Marathon	Arizona	-9.9	-19.9	-17.9	+29.1	5
Hokkaido Marathon	Japan	-9.8	-19.5	-17.6	—	10
Abingdon Marathon	United Kingdom	-9.5	-19.0	-17.1	—	9
REVEL MT Charleston	Nevada	-8.5	-17.1	-15.4	+21.8	10
Adelaide Festival City Marathon	Australia	-8.5	-17.0	-15.3	—	23
Spring Chance Bq.2 Marathon	Illinois	-8.3	-16.6	-15.0	+10.3	6
Tunnel Lite Marathon	Nevada	-7.8	-15.6	-14.0	+19.1	11
Last Chance Bq.2 Chicagoland Marathon	Illinois	-7.7	-15.3	-13.8	+12.3	9
REVEL Big Bear and Half Marathon	California	-7.6	-15.1	-13.6	+21.7	8
Mountain To Surf Marathon	Czech Republic	-7.5	-15.1	-13.5	—	12
Last Chance Bq.2 Grand Rapids Marathon	Michigan	-7.2	-14.5	-13.0	+10.0	6
City Of Christchurch Marathon	New Zealand	-6.9	-13.9	-12.5	—	22
Light At The End Of The Tunnel Marathon	Washington	-6.4	-12.8	-11.5	+15.4	11
Canberra Marathon	Australia	-5.8	-11.6	-10.4	—	22
MT. Nebo Marathon	Utah	-5.7	-11.4	-10.3	+13.4	10
Carmel Marathon	Indiana	-5.6	-11.2	-10.1	+12.1	13
Hudson Mohawk Marathon	New York	-5.6	-11.2	-10.1	+14.9	10
Sunshine Coast Marathon	Australia	-5.5	-11.0	-9.9	—	9
Melbourne Marathon	Australia	-5.5	-11.0	-9.9	—	22
Townsville Marathon	Australia	-5.5	-11.0	-9.9	—	16
Newport Wales Marathon	United Kingdom	-5.5	-10.9	-9.8	—	5
Gold Coast Airport Marathon	Australia	-5.1	-10.3	-9.3	—	20
Brisbane Marathon	Australia	-5.0	-10.0	-9.0	—	10
Perth Marathon	Australia	-5.0	-10.1	-9.1	—	21
<i>10 Slowest Courses</i>						
Great Wall Marathon	United States	+37.2	+74.4	+66.9	—	9
Greenway Trail Marathon	Maryland	+38.3	+76.6	+68.9	-67.3	10
Bataan Memorial Death March	New Mexico	+39.4	+78.9	+71.0	-49.2	22
Aspen Backcountry Marathon	Colorado	+40.6	+81.2	+73.1	—	5
Six Foot Track (Ultra) Marathon	Australia	+42.8	+85.6	+77.1	—	6
Moab Trail Marathon	Utah	+42.9	+85.7	+77.2	-84.0	11
The North Face Endurance Challenge - Bear Mountain	New York	+46.5	+92.9	+83.6	—	9
Moose Mountain Marathon	Minnesota	+48.0	+95.9	+86.3	—	13
Leadville Trail Marathon	Colorado	+54.5	+109.1	+98.2	-103.8	22
Pikes Peak Marathon	Colorado	+76.0	+152.1	+136.9	-122.4	24